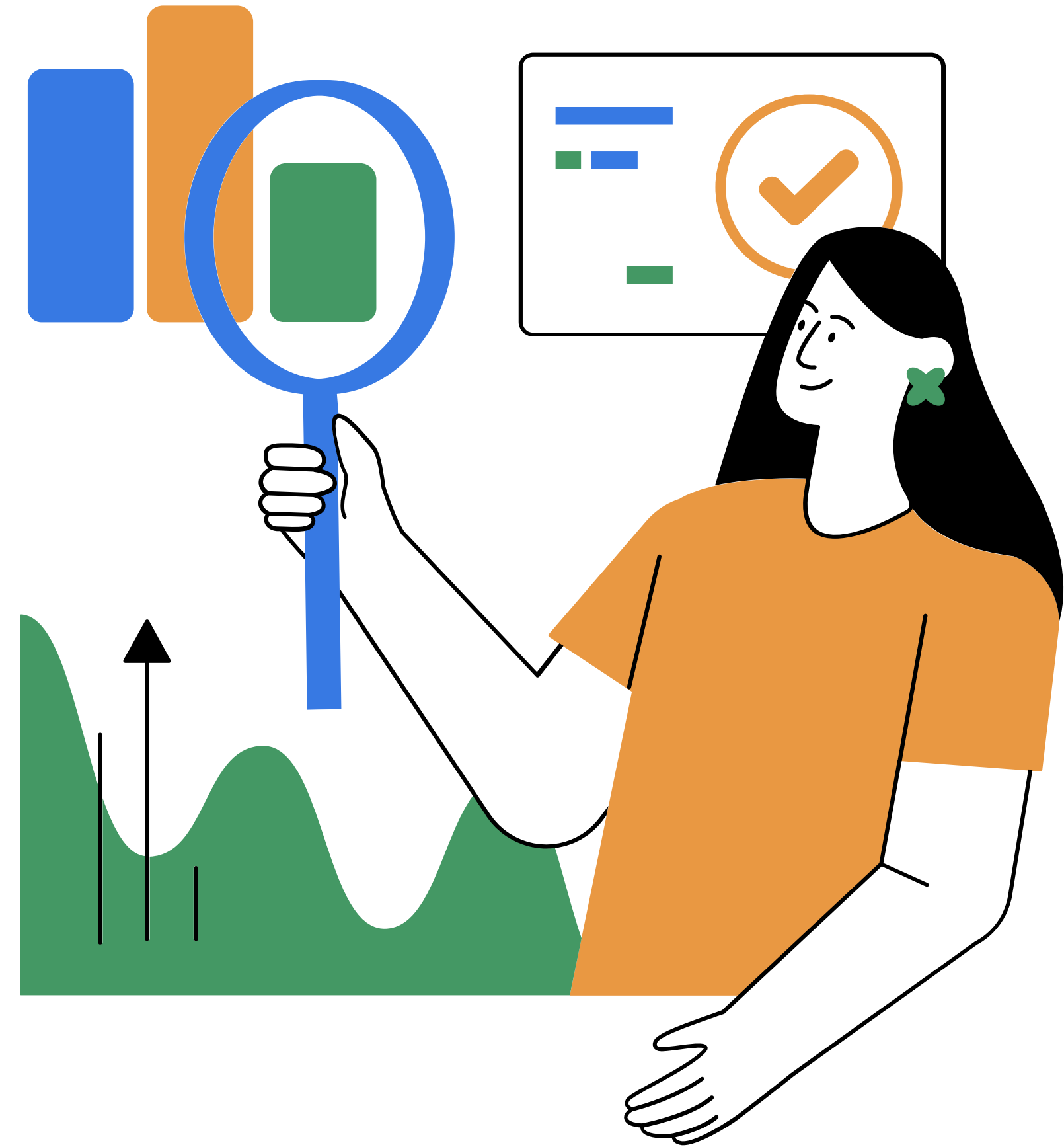


# Global Public Health Indicators

**Presented by Truth Seekers**

**Data Mining Project**



# Introduction

**Global Public Health Indicators are measurable values that help us understand the overall health status of populations around the world. These indicators allow countries and organizations to track progress, compare health outcomes, and plan policies for improving health. Common indicators include life expectancy, infant mortality, immunization coverage, maternal health, access to healthcare, and disease burden.**



# Motivation

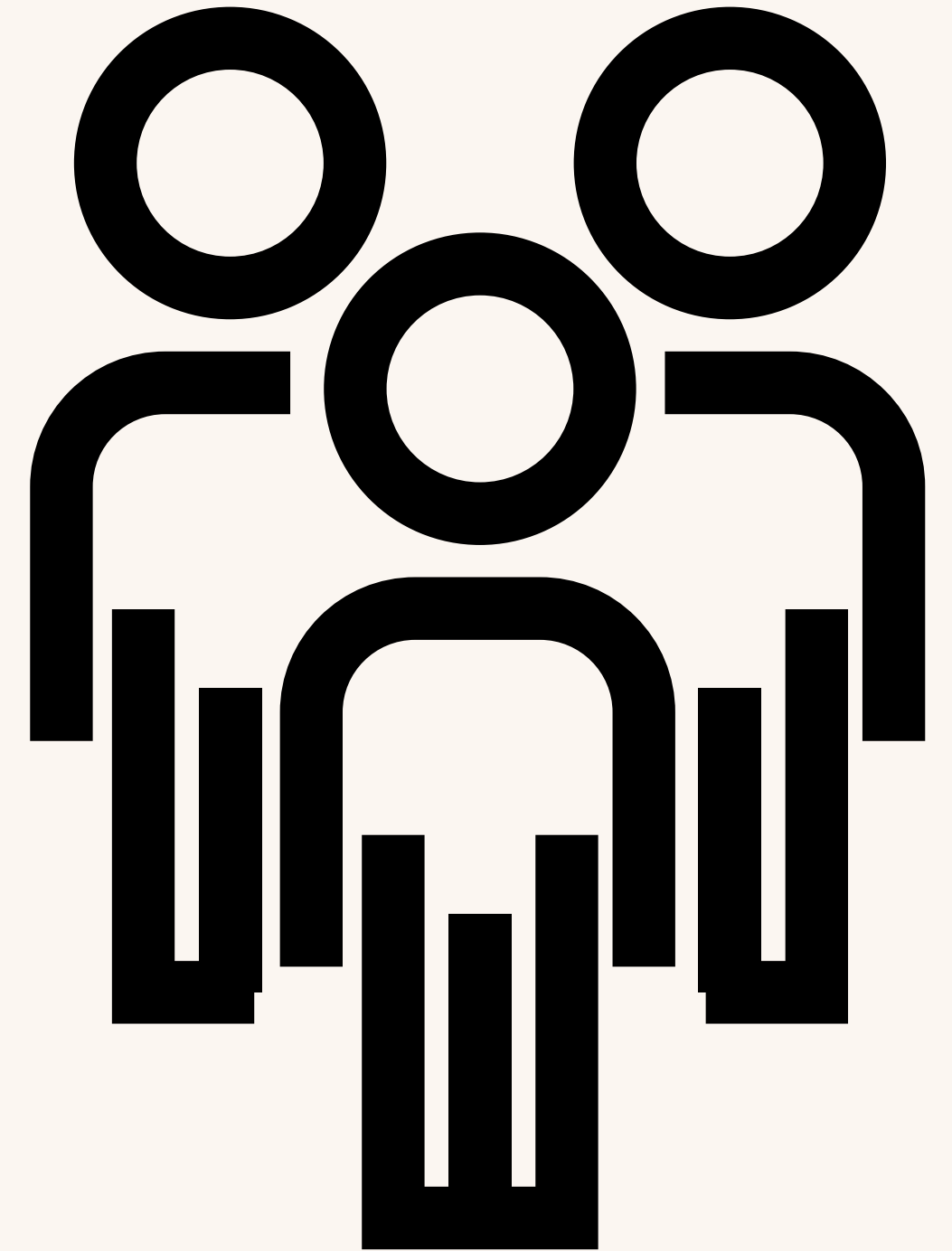
**Why this topic is important:**

- **To understand how healthy or unhealthy populations across the world are**
- **To identify health inequalities between developed and developing nations**
- **To measure progress toward global health goals (like SDG-3: Good Health and Well-Being)**
- **To learn how global organizations (WHO, UNICEF, World Bank) monitor health trends**
- **Because global diseases (COVID-19, HIV/AIDS, TB) affect everyone**



# Our Team:

1. Uday Kumar Choudhary
2. Ayush Kumar Pandey
3. Nakul Sharma
4. Anuj Chhajed



# UDAY'S WORK



# \* Insights on Life Expectancy Drivers

```
▶ import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

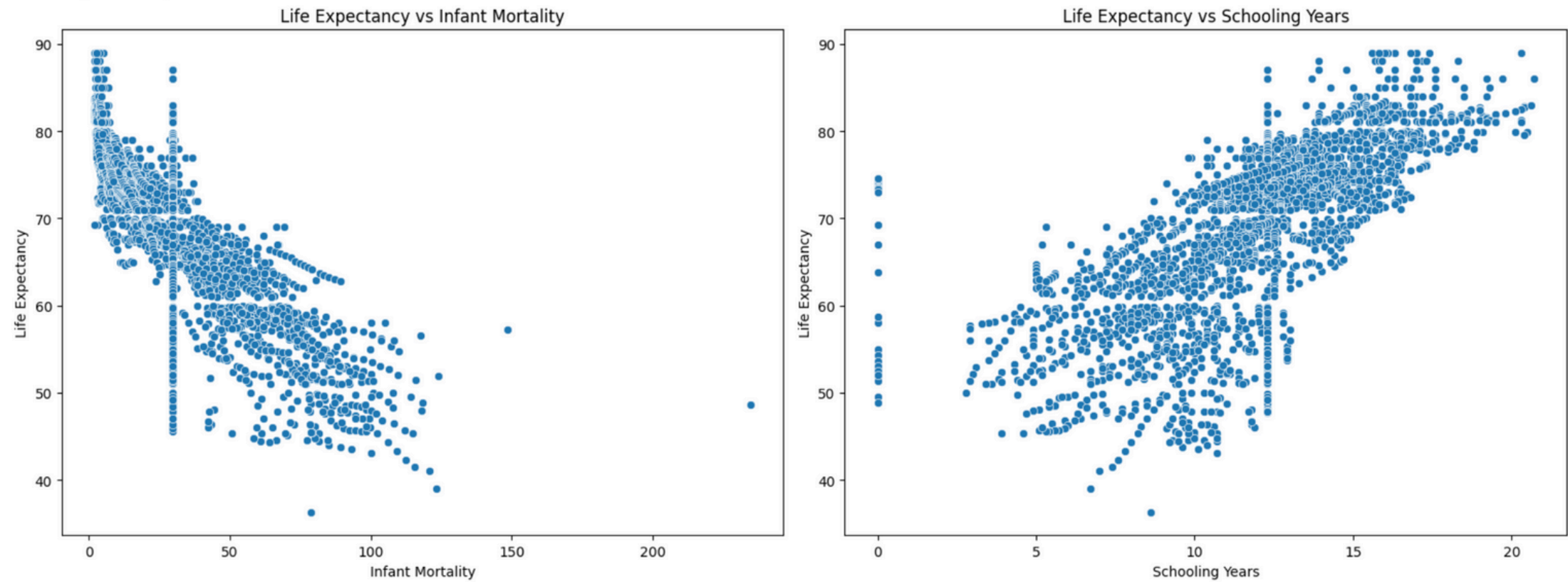
target_variable = 'life_expectancy_'
predictor_variables = ['gdp', 'schooling', 'adult_mortality', 'infant_mortality']

correlations = df[predictor_variables + [target_variable]].corr()[target_variable]

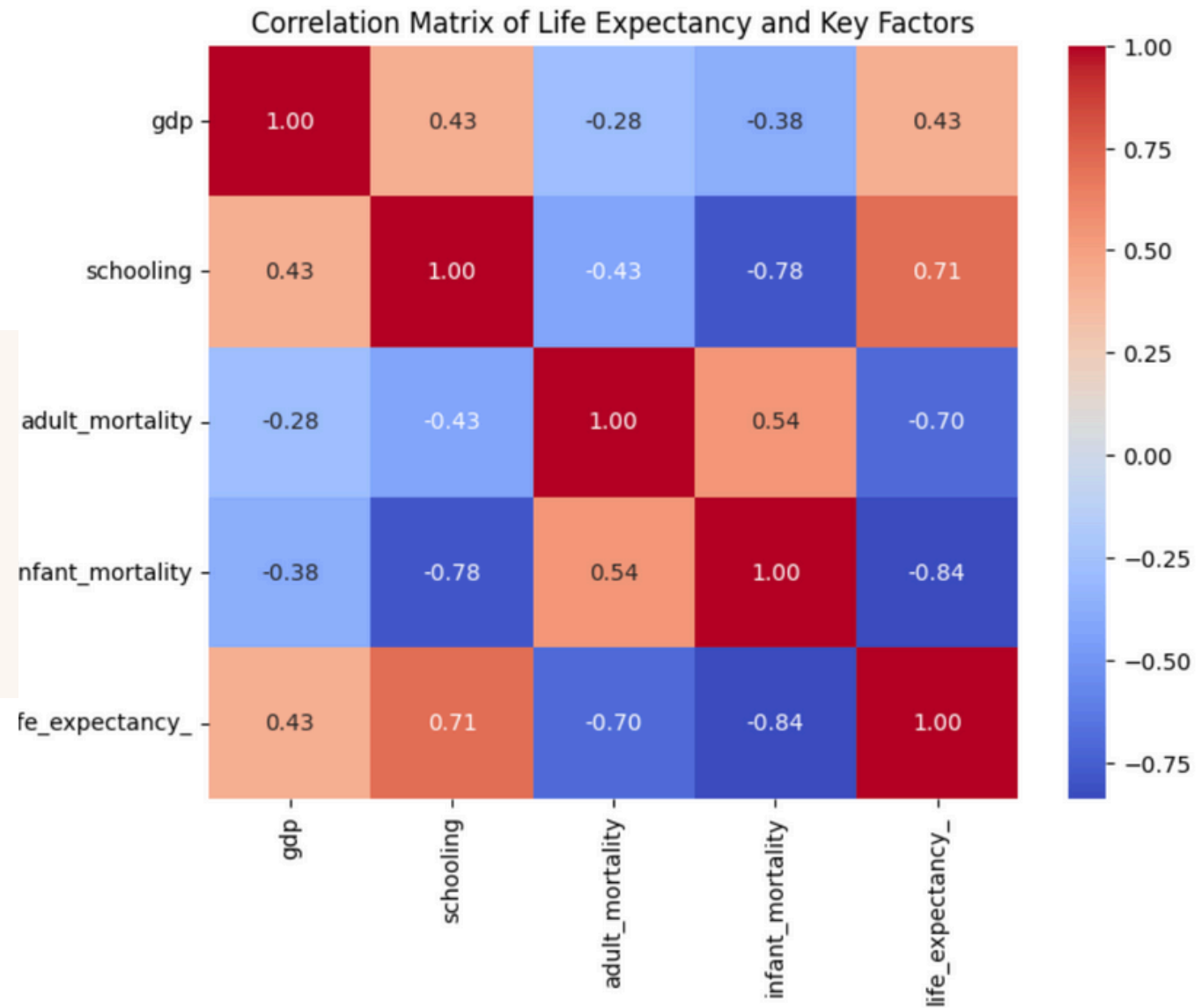
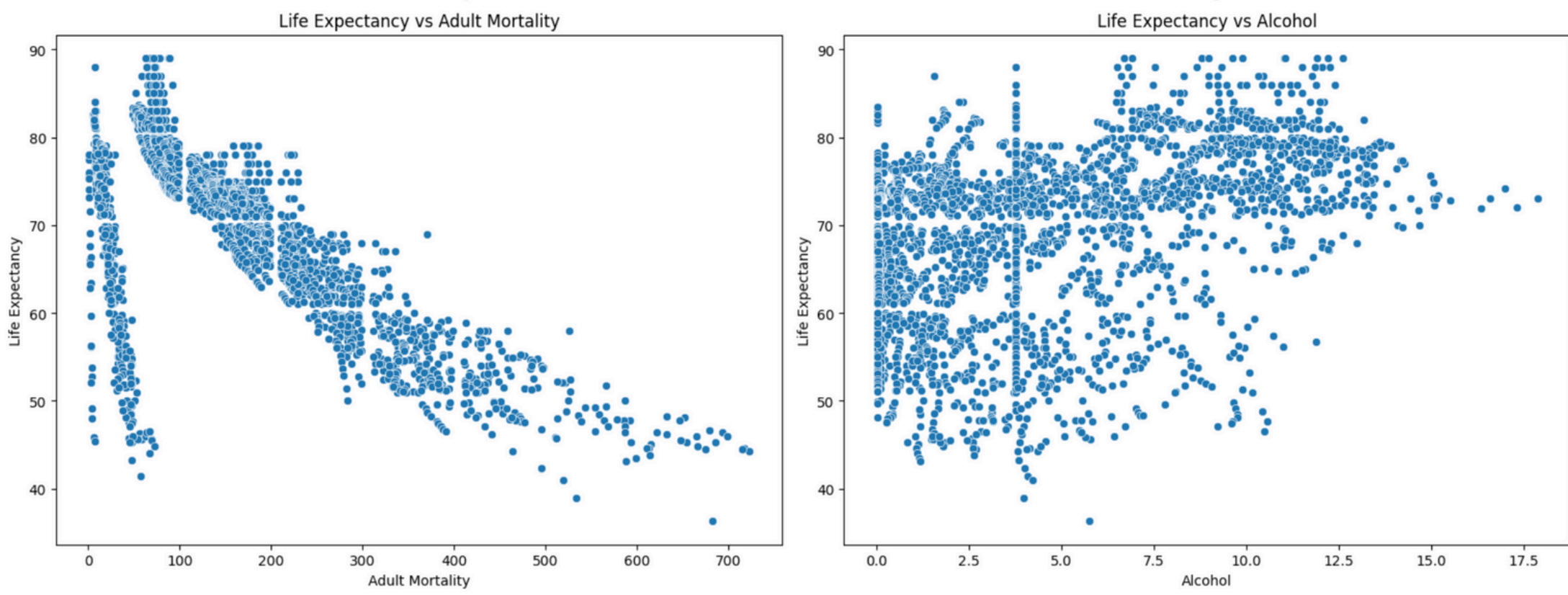
print("Pearson Correlation Coefficients with Life Expectancy:")
print(correlations.drop(target_variable))
```

```
... Pearson Correlation Coefficients with Life Expectancy:
gdp          0.430895
schooling    0.713738
adult_mortality -0.696359
infant_mortality -0.838497
Name: life_expectancy_, dtype: float64
```

Generating scatter plots...



# ScatterPlot & Heatmaps



relation matrix heatman generated.

# \* Insights on Life Expectancy Drivers

## 1. Strongest Correlates of Life Expectancy

- Infant Mortality: strongest negative correlation (-0.84)
- Schooling Years: strongest positive correlation (+0.71)
- Adult Mortality: strong negative (-0.70)
- GDP: moderate positive (+0.43)
- → Lower infant mortality and higher schooling are the biggest drivers of higher life expectancy.

## 2. Does Higher GDP Increase Life Expectancy?

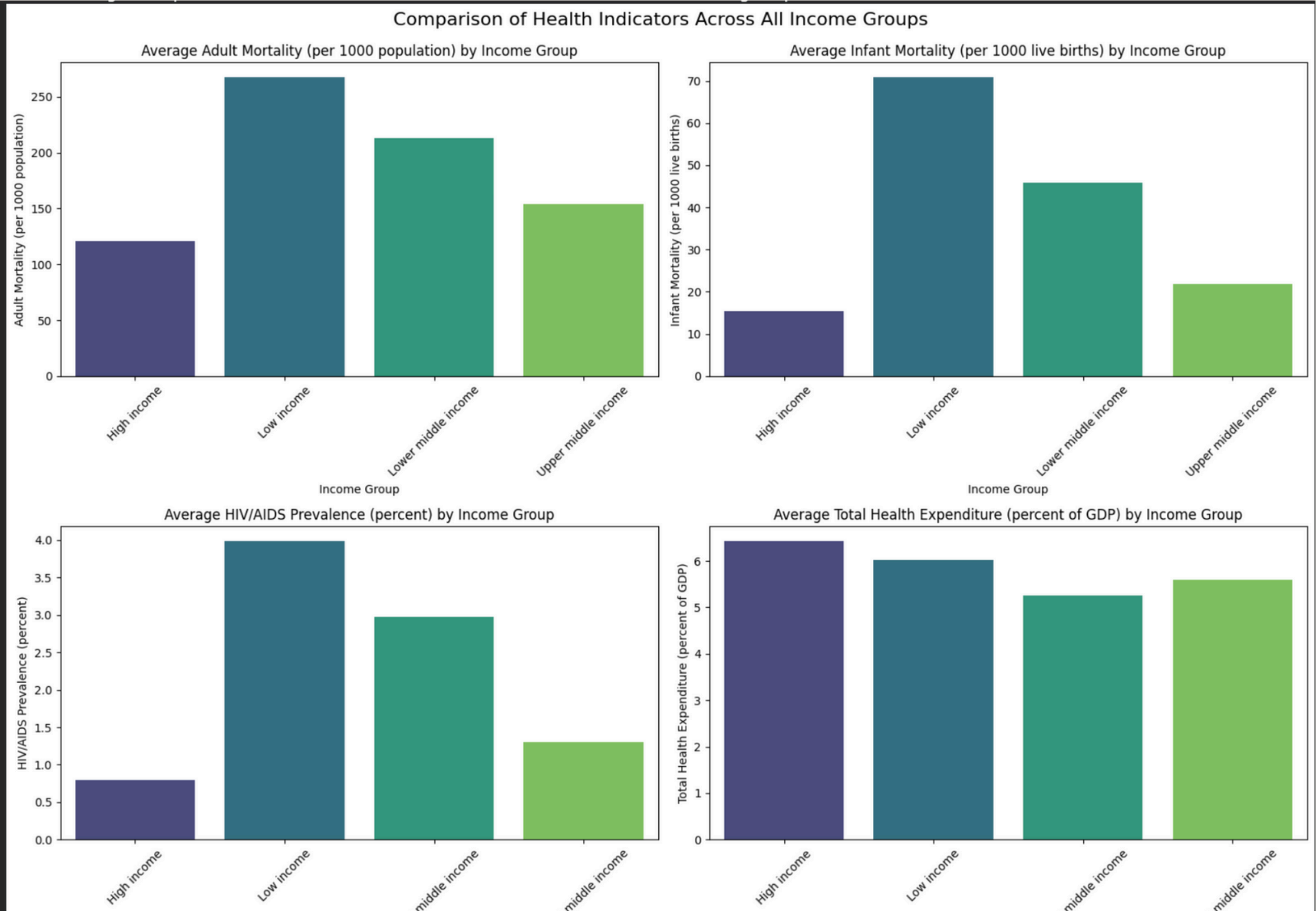
- Yes — moderate positive correlation (+0.43).
- Higher GDP generally aligns with higher life expectancy, but the relationship is weaker than schooling or mortality.

## 3. Impact of Schooling Years

- Strong positive correlation (+0.71).
- More schooling → better health awareness & outcomes → higher life expectancy.



# Comparison Of Health Indicator With Income Groups



# \* Mortality & Disease Burden

## Adult Mortality

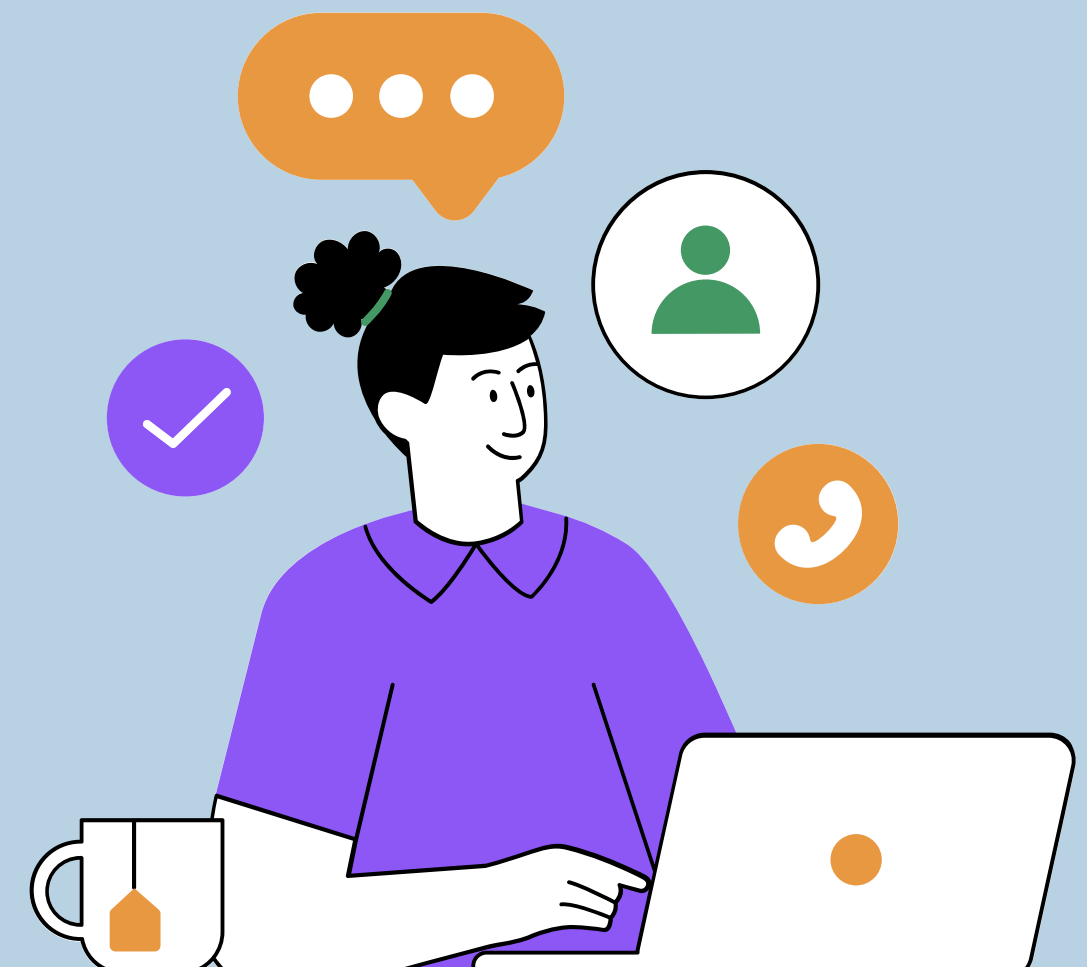
- High Income: Lowest (120.81 per 1000)
- Upper Middle: 154.14
- Lower Middle: 212.72
- Low Income: Highest (267.21)
- Clear inverse relationship between income and adult mortality.

## Infant Mortality

- High Income: Lowest (15.33 per 1000 live births)
- Upper Middle: 21.79
- Lower Middle: 45.97
- Low Income: Highest (70.80)
- Economic prosperity strongly boosts child survival.

## HIV/AIDS Prevalence

- High Income: Lowest (0.80%)
- Upper Middle: 1.31%
- Lower Middle: 2.98%
- Low Income: Highest (3.99%)
- Lower income → higher disease burden.



# \* Key Insights & Health Expenditure

## Income-Health Correlation

- Mortality doubles to triples from high to low income.
- Strong inverse relationship: higher income = lower mortality.

## Health Expenditure (% of GDP)

- High Income: 6.43% | Upper Middle: 5.59%
- Lower Middle: 5.25% | Low Income: 6.02%
- **Paradox: Spending volume  $\neq$  better outcomes; efficiency matters.**

## Critical Takeaway

- Socio-economic development is the primary health determinant.
- Income level influences child survival, disease prevalence, and infrastructure.
- Beyond spending: governance, access, and system effectiveness are crucial.

# \* Identify and list the top 10 countries with the highest infant mortality.

```
print("\n" + "-" * 30 + "\n")
print("Identifying top 10 countries with highest infant mortality rates...")

country_infant_mortality = df.groupby('country')['infant_mortality'].mean()

top_10_infant_mortality = country_infant_mortality.sort_values(ascending=False).head(10)

print("Top 10 Countries with Highest Average Infant Mortality Rates:")
print(top_10_infant_mortality)

print("\n" + "-" * 30 + "\n")
```

```
-----

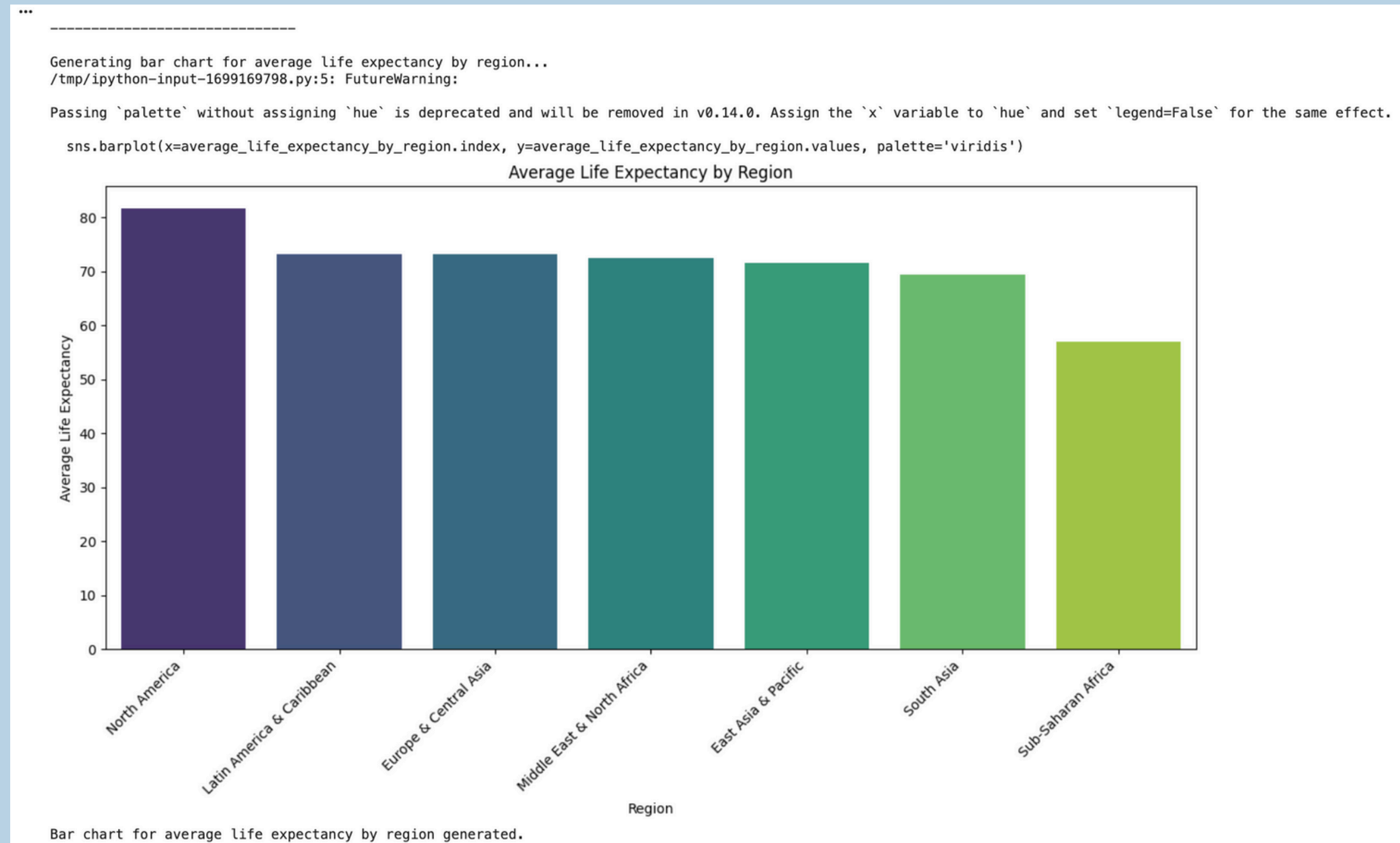
Identifying top 10 countries with highest infant mortality rates...
Top 10 Countries with Highest Average Infant Mortality Rates:
country
Sierra Leone      99.80625
South Sudan        97.62500
Central African Republic  97.53750
Chad                87.71250
Afghanistan         85.87500
Liberia             83.95000
Mali                83.66875
Equatorial Guinea   83.50000
Angola              82.06875
Niger               81.31250
Name: infant_mortality, dtype: float64

-----
```

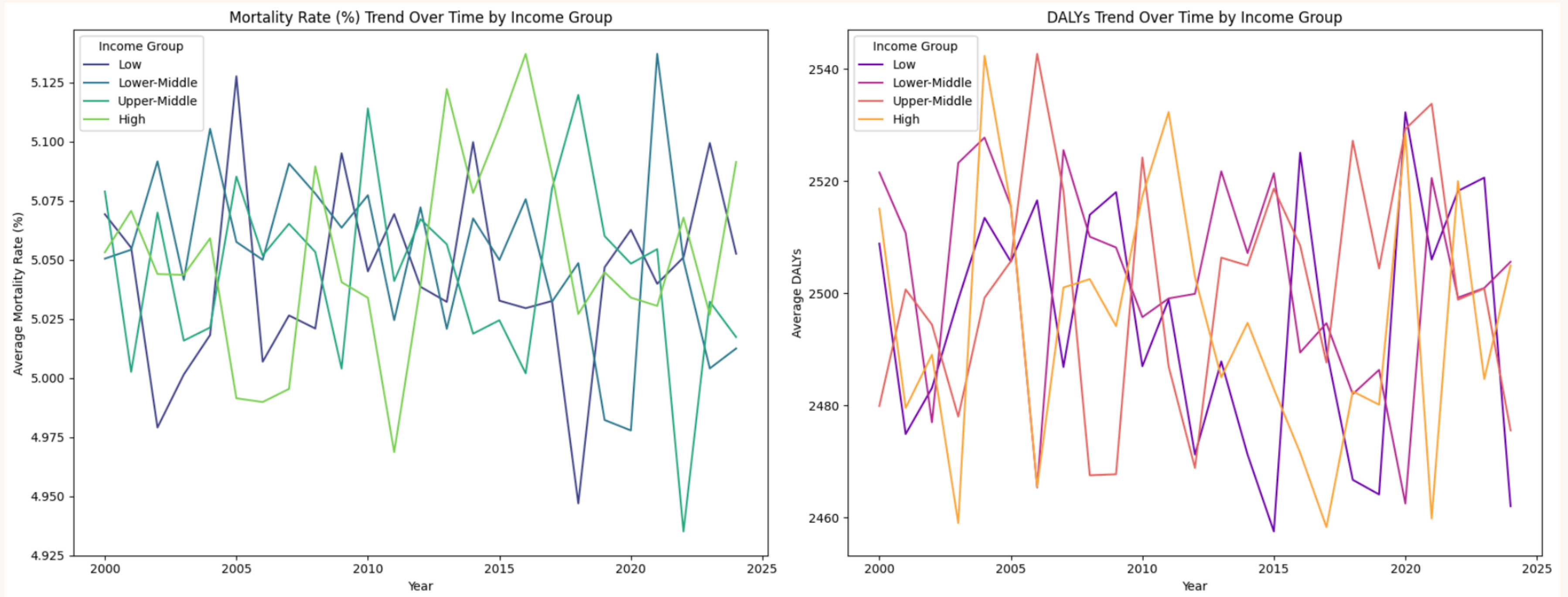
## Why These Countries Have the Highest Infant Mortality Rates??

These countries share structural issues such as weak healthcare systems, extreme poverty, political instability, lack of maternal healthcare, poor sanitation, and high prevalence of preventable diseases. Together, these factors limit access to essential newborn care, leading to very high infant mortality rates compared to the rest of the world.

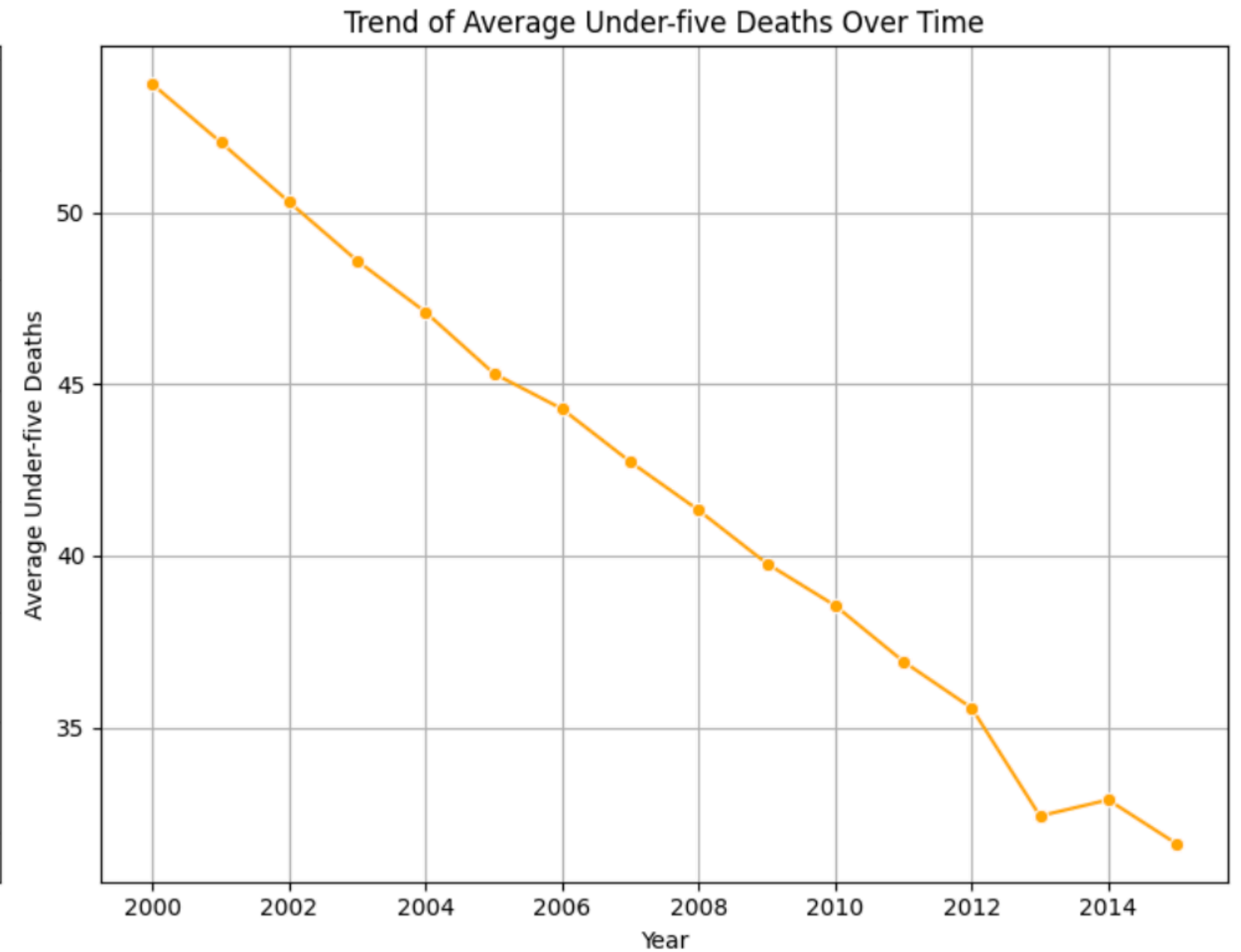
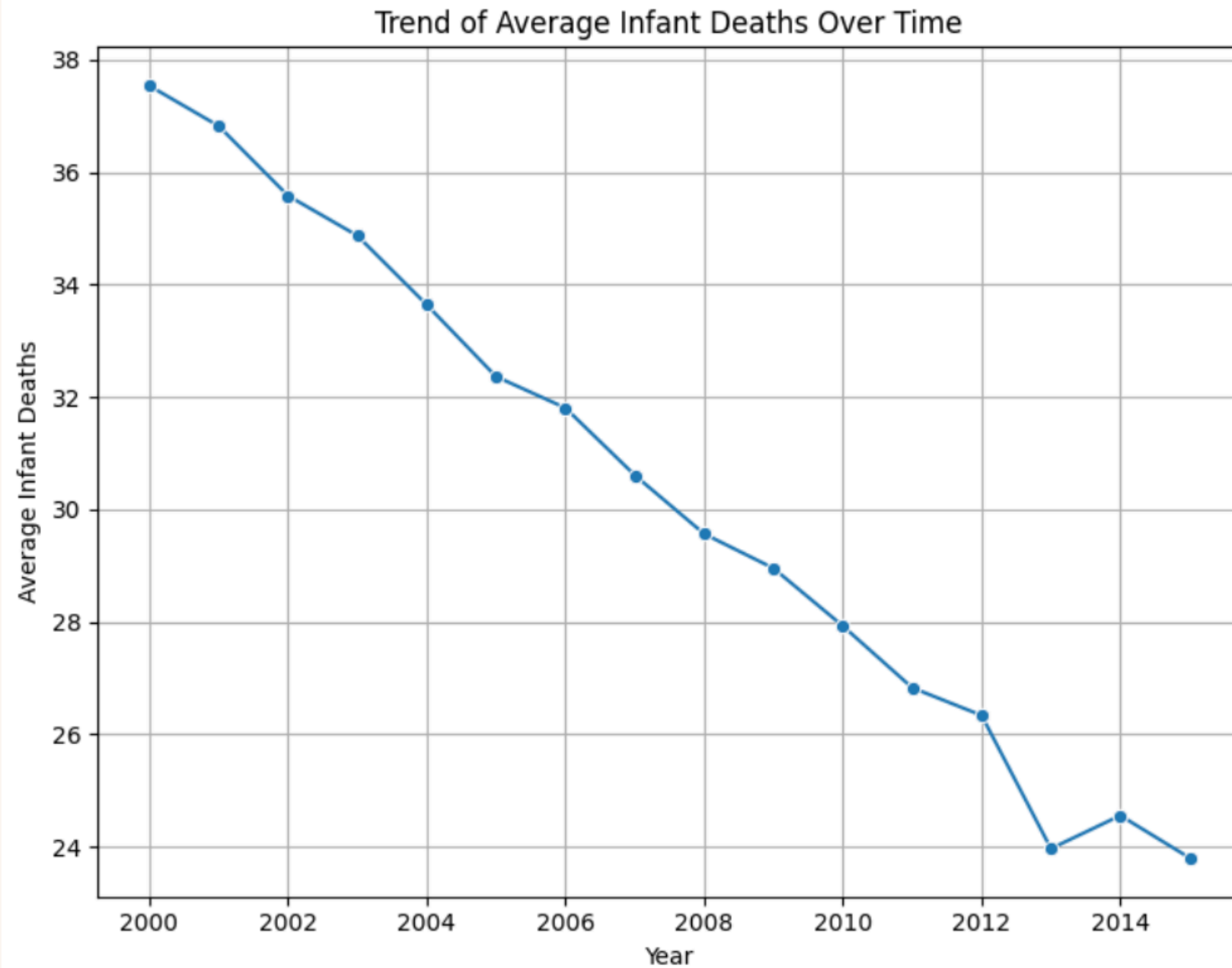
# \* Bar chart for average life expectancy by region generated.



# \* Trend Over Time



# \* Trend Over Time



# \* Definitions & Mortality Trend:

## **Mortality Rate (%)**

- The percentage of people who die in a population within a specific period.

## **DALYs (Disability-Adjusted Life Years)**

- A measure of total health loss = Years lost due to early death + Years lived with illness/disability.
- Higher DALYs → Worse health outcomes.

## **Mortality Rate Trend (2000–2024)**

- High-income countries always show lower mortality rates.
- Low-income countries consistently show higher mortality rates.
- The gap between income groups remains almost unchanged, showing persistent health inequality.



# \* DALYs Trend & Overall Insight:

## **DALYs Trend (2000–2024)**

- Low-income and lower-middle-income groups have a higher disease burden (higher DALYs).
- High-income countries maintain lower DALYs, meaning better health and healthcare access.
- The difference between income groups stays wide over time.

## **Overall Insight**

- Richer countries = Better health outcomes (lower mortality + lower DALYs).
- Poorer countries = Higher mortality + greater years lost to illness.
- The health inequality gap between income groups has not reduced significantly from 2000–2024.



# AYUSH 'S WORK





# Analyzing the Impact of Vaccination on Infant Mortality

- **Objective:** To examine global measles vaccination coverage and infant mortality rates and understand how vaccination influences infant survival.
- **Analysis Scope:** Country-level data from 1980 to 2024, focusing on recent decades (2010+) for policy relevance.
- **Data Source:** WHO and global health datasets

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
vax_df = pd.read_csv("/content/global-vaccination-coverage.csv")
mort_df = pd.read_csv("/content/infant_mortality .csv")
```

```
# Check missing data percentage
percentage_missing_values =(df.isnull().mean() * 100).round(2)
print("percentage missing values", percentage_missing_values)
```

```
percentage missing values Country
ISO_Code 11.18
Year 0.00
HepB_3dose 38.67
Hib_3dose 48.23
Polio_1dose 77.65
Measles_1dose 1.68
Pneumo_3dose 75.44
Polio_3dose 0.26
Rubella_1dose 40.44
Rotavirus_1dose 81.32
DTP_3dose 0.32
dtype: float64
```

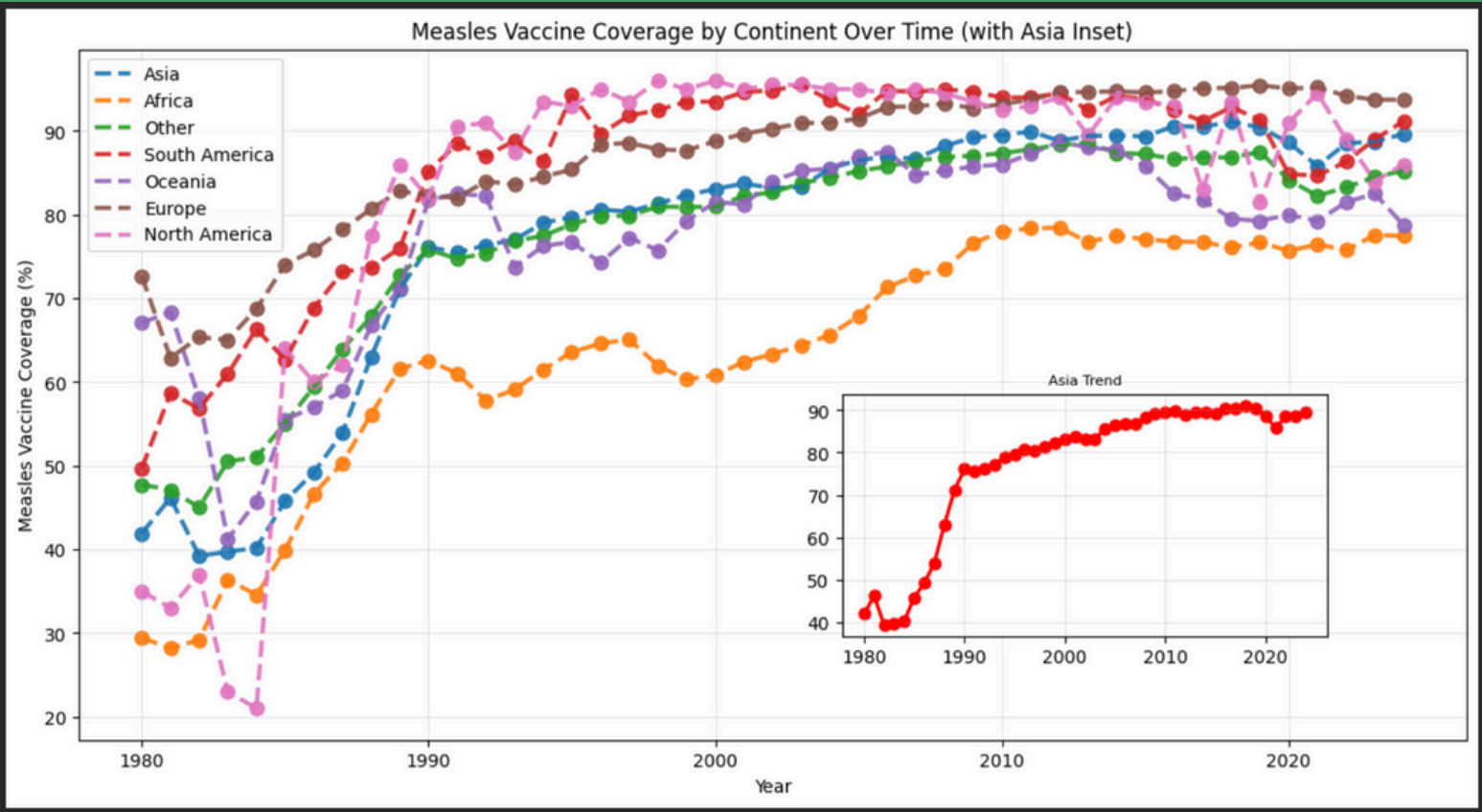
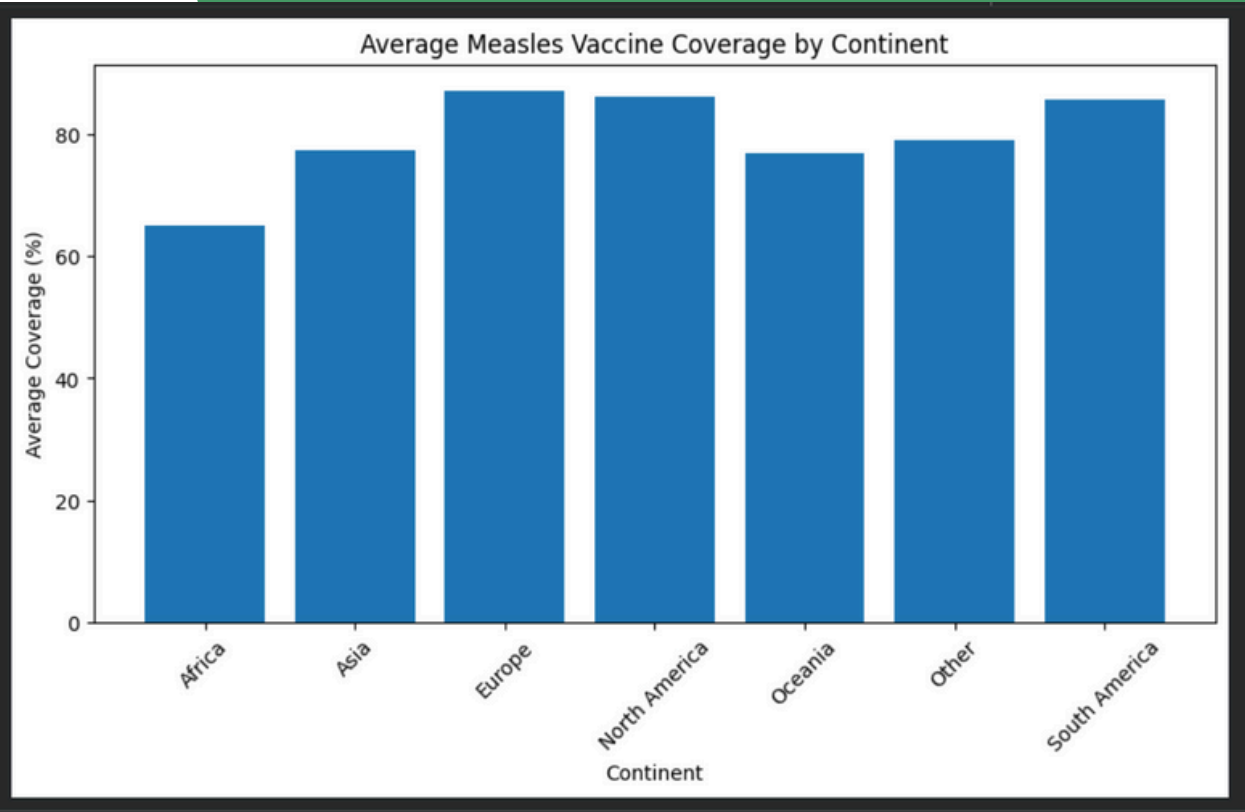
|                 |      |
|-----------------|------|
|                 | 0    |
| Country         | 0    |
| ISO_Code        | 1029 |
| Year            | 0    |
| HepB_3dose      | 3559 |
| Hib_3dose       | 4439 |
| Polio_1dose     | 7146 |
| Measles_1dose   | 155  |
| Pneumo_3dose    | 6943 |
| Polio_3dose     | 24   |
| Rubella_1dose   | 3722 |
| Rotavirus_1dose | 7484 |
| DTP_3dose       | 29   |
| dtype: int64    |      |





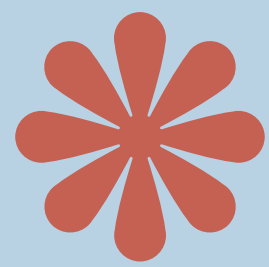
# Global Measles Vaccination Coverage Trends

- Dataset contains measles vaccine coverage percentages for one-year-old children across countries and years.



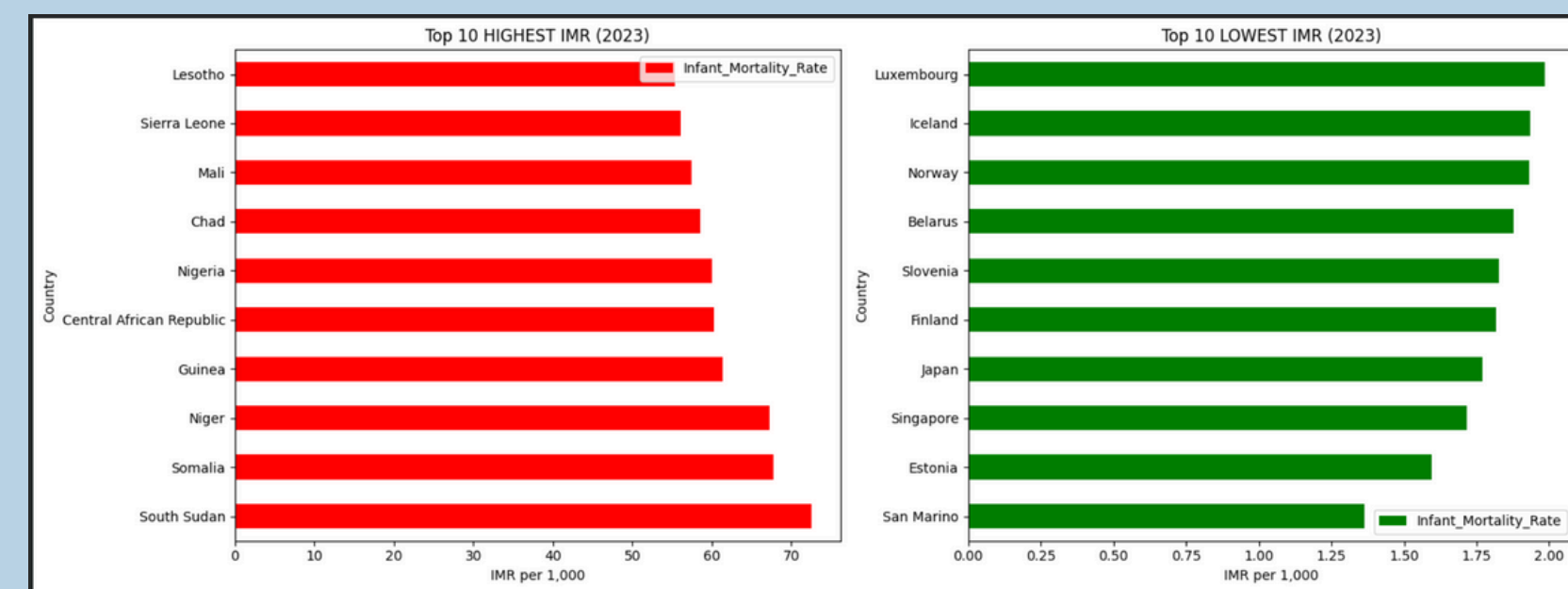
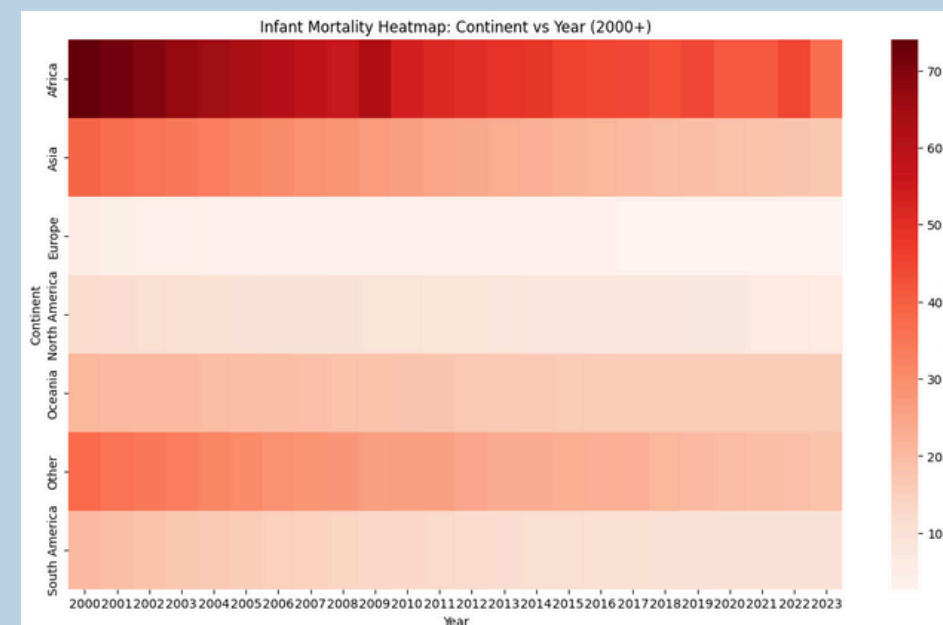
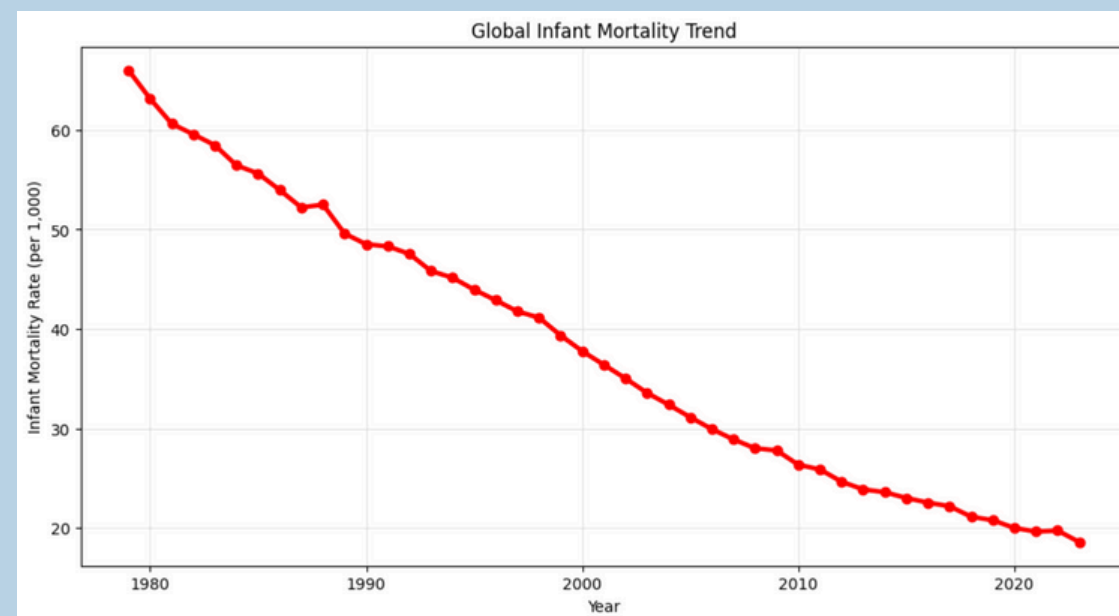
• **Key Observation:** Measles vaccination coverage has steadily improved worldwide, with progress varying by continent and country.

|    | Country     | ISO_Code | Year | Measles_Vaccine_Coverage |
|----|-------------|----------|------|--------------------------|
| 0  | Afghanistan | AFG      | 2007 | 55.000000                |
| 1  | Afghanistan | AFG      | 2008 | 59.000000                |
| 2  | Afghanistan | AFG      | 2009 | 60.000000                |
| 3  | Afghanistan | AFG      | 2010 | 62.000000                |
| 4  | Afghanistan | AFG      | 2011 | 64.000000                |
| 5  | Afghanistan | AFG      | 2012 | 59.000000                |
| 6  | Afghanistan | AFG      | 2013 | 57.000000                |
| 7  | Afghanistan | AFG      | 2014 | 60.000000                |
| 8  | Afghanistan | AFG      | 2015 | 62.000000                |
| 9  | Afghanistan | AFG      | 2016 | 64.000000                |
| 10 | Afghanistan | AFG      | 2017 | 64.000000                |
| 11 | Afghanistan | AFG      | 2018 | 66.000000                |
| 12 | Afghanistan | AFG      | 2019 | 57.000000                |
| 13 | Afghanistan | AFG      | 2020 | 57.000000                |
| 14 | Afghanistan | AFG      | 2021 | 51.000000                |
| 15 | Afghanistan | AFG      | 2022 | 56.000000                |
| 16 | Afghanistan | AFG      | 2023 | 55.000000                |
| 17 | Afghanistan | AFG      | 2024 | 55.000000                |
| 18 | Afghanistan | AFG      | 1980 | 11.000000                |
| 19 | Afghanistan | AFG      | 1982 | 8.000000                 |
| 20 | Afghanistan | AFG      | 1983 | 9.000000                 |
| 21 | Afghanistan | AFG      | 1984 | 14.000000                |
| 22 | Afghanistan | AFG      | 1985 | 14.000000                |
| 23 | Afghanistan | AFG      | 1986 | 14.000000                |
| 24 | Afghanistan | AFG      | 1987 | 31.000000                |
| 25 | Afghanistan | AFG      | 1988 | 34.000000                |
| 26 | Afghanistan | AFG      | 1989 | 22.000000                |
| 27 | Afghanistan | AFG      | 1990 | 20.000000                |
| 28 | Afghanistan | AFG      | 1991 | 19.000000                |

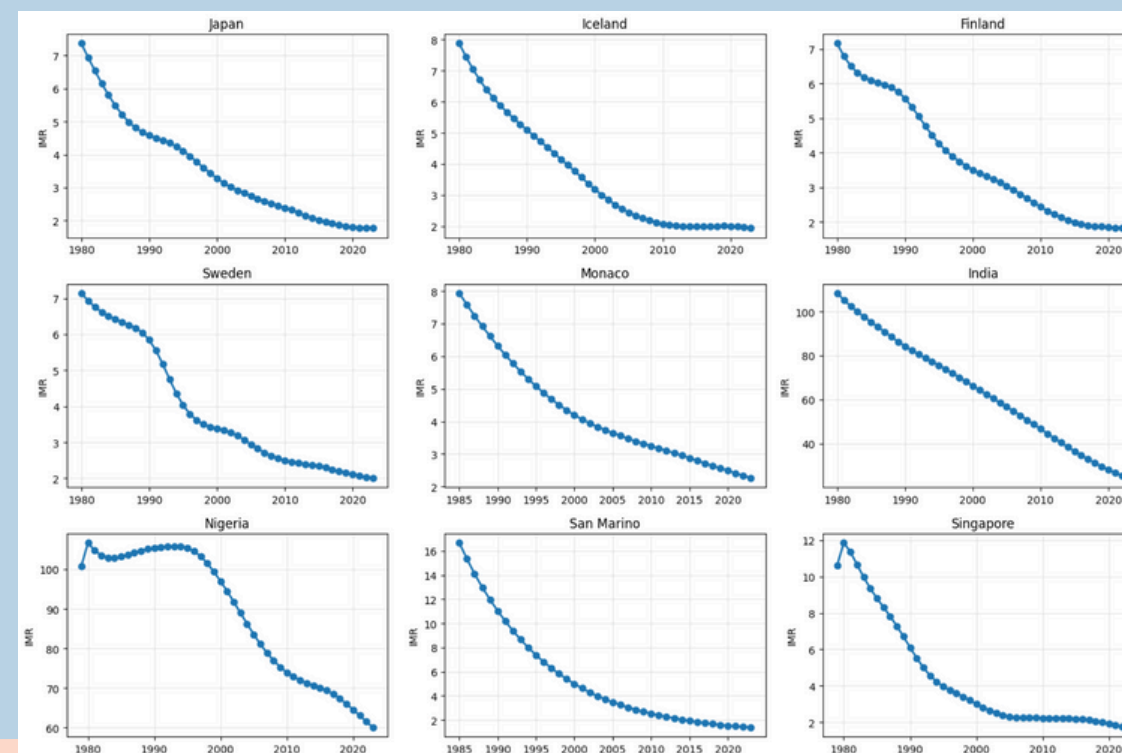


# Infant Mortality Rates Across the World

- Dataset provides infant mortality rates (deaths per 1,000 live births) by country and year.



|   | Continent     | Reduction_% |
|---|---------------|-------------|
| 0 | Asia          | 55.5        |
| 3 | South America | 53.1        |
| 1 | Other         | 50.7        |
| 2 | Africa        | 49.9        |
| 6 | North America | 42.8        |
| 5 | Europe        | 42.6        |
| 4 | Oceania       | 24.9        |

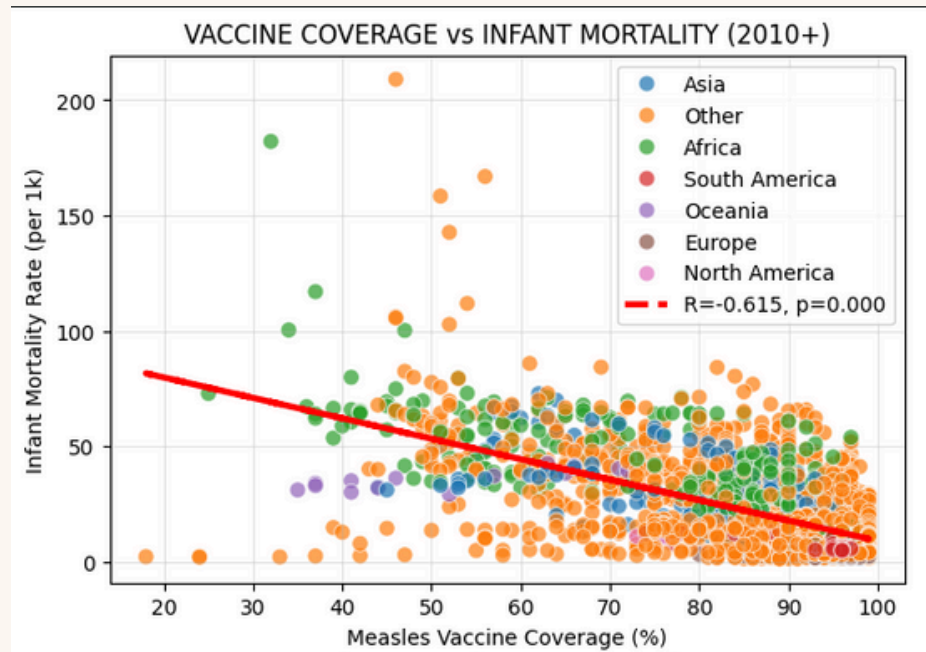


| Continent Mortality Ranking: |      |
|------------------------------|------|
| Continent                    |      |
| Africa                       | 70.3 |
| Asia                         | 40.8 |
| Other                        | 36.4 |
| South America                | 23.1 |
| Oceania                      | 21.8 |
| North America                | 13.4 |
| Europe                       | 5.8  |
| Name: Infant_Mortality_Rate, |      |

- Visualisation** : Highlights declining infant mortality globally but persistent high rates in parts of Africa and some Asian countries.
- Key Observation**: Despite overall decline, infant mortality rates remain disproportionately high in lower-income regions.

# CORRELATION BETWEEN VACCINATION AND INFANT MORTALITY

- Merged vaccination and mortality datasets to analyze relationship across countries and years.

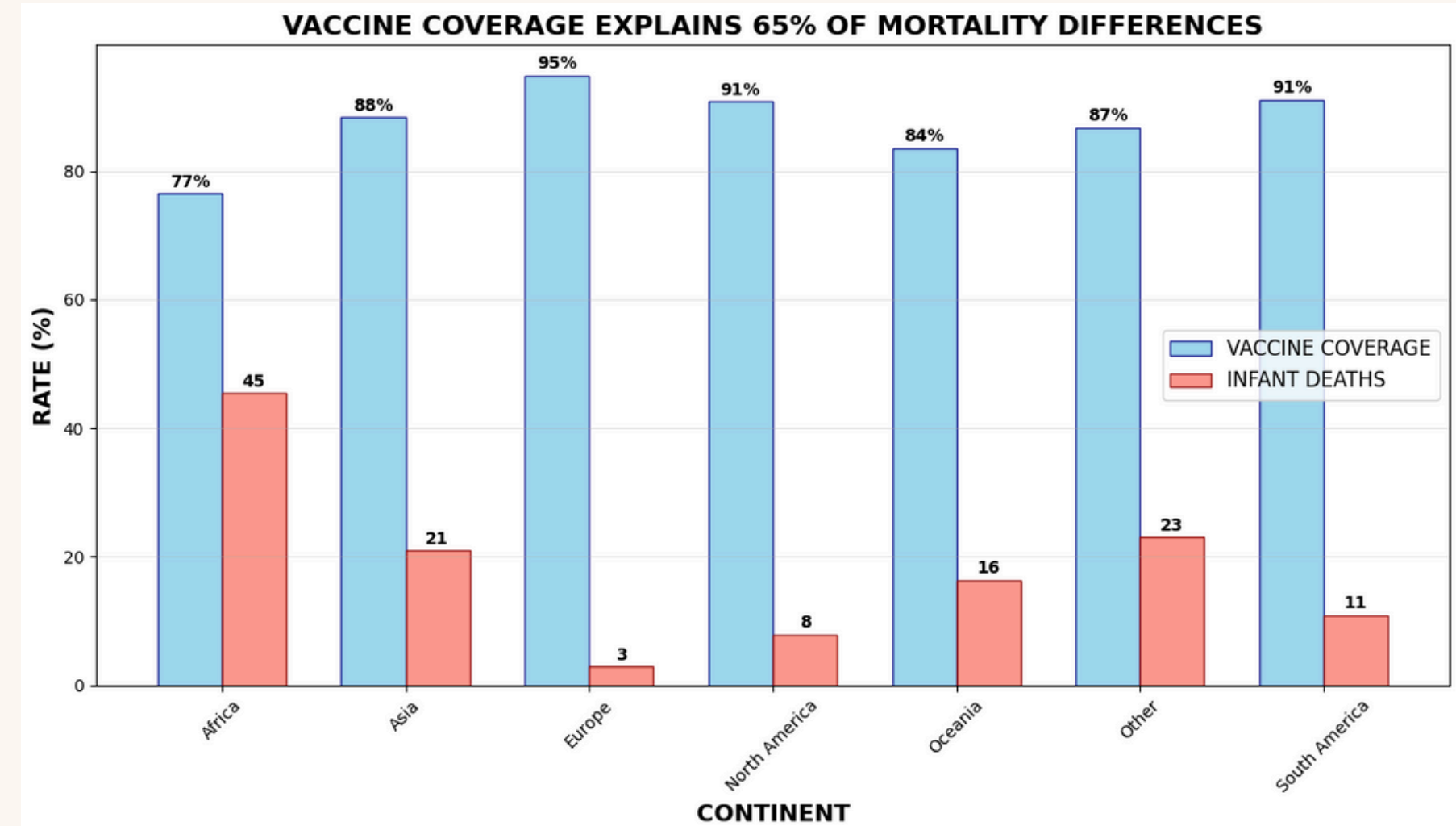
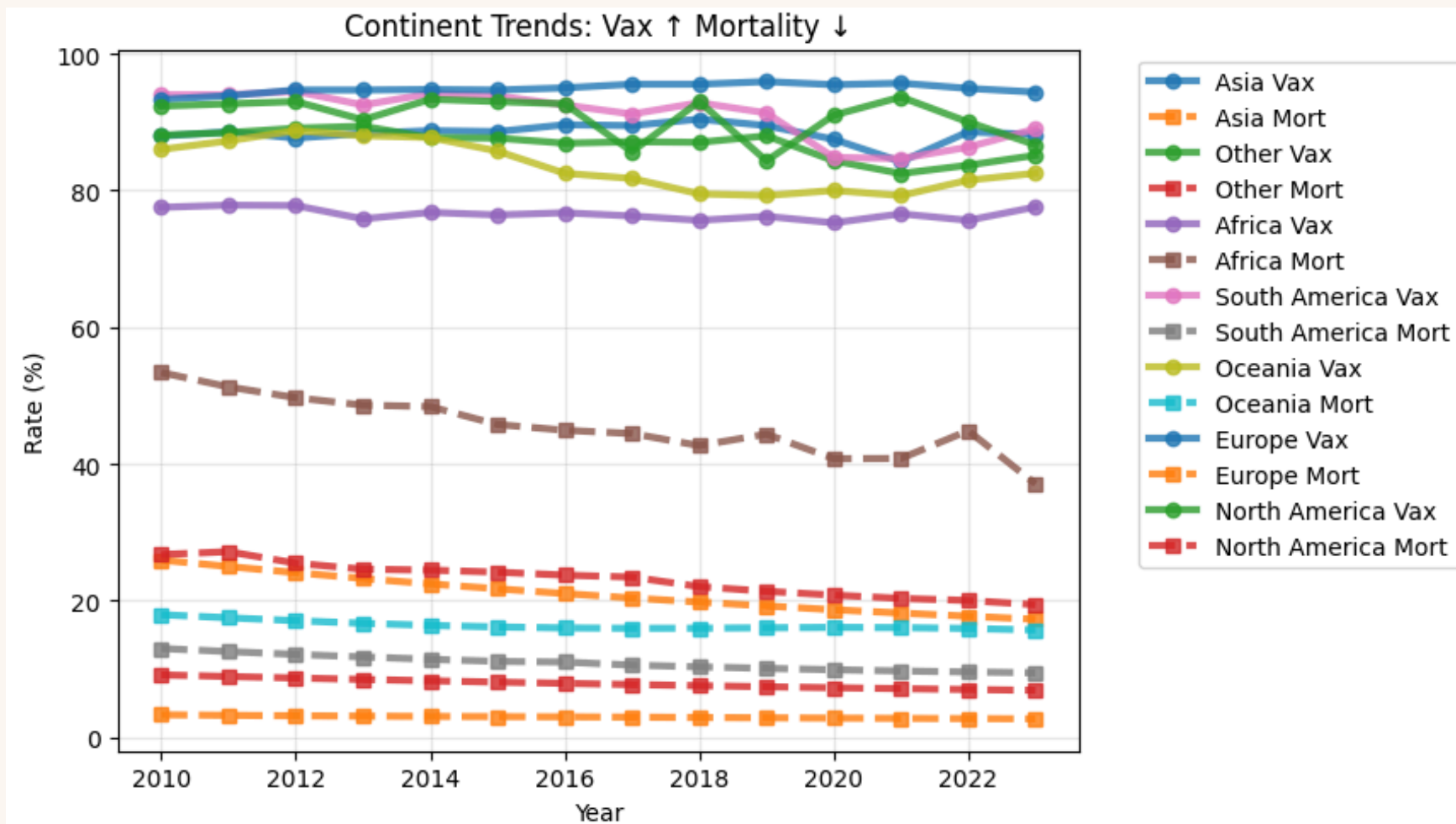


## STATISTICS FROM ANALYSIS:

$R = -0.617 \rightarrow$  **STRONG negative relationship**  
(Higher vaccine % = Lower infant deaths)

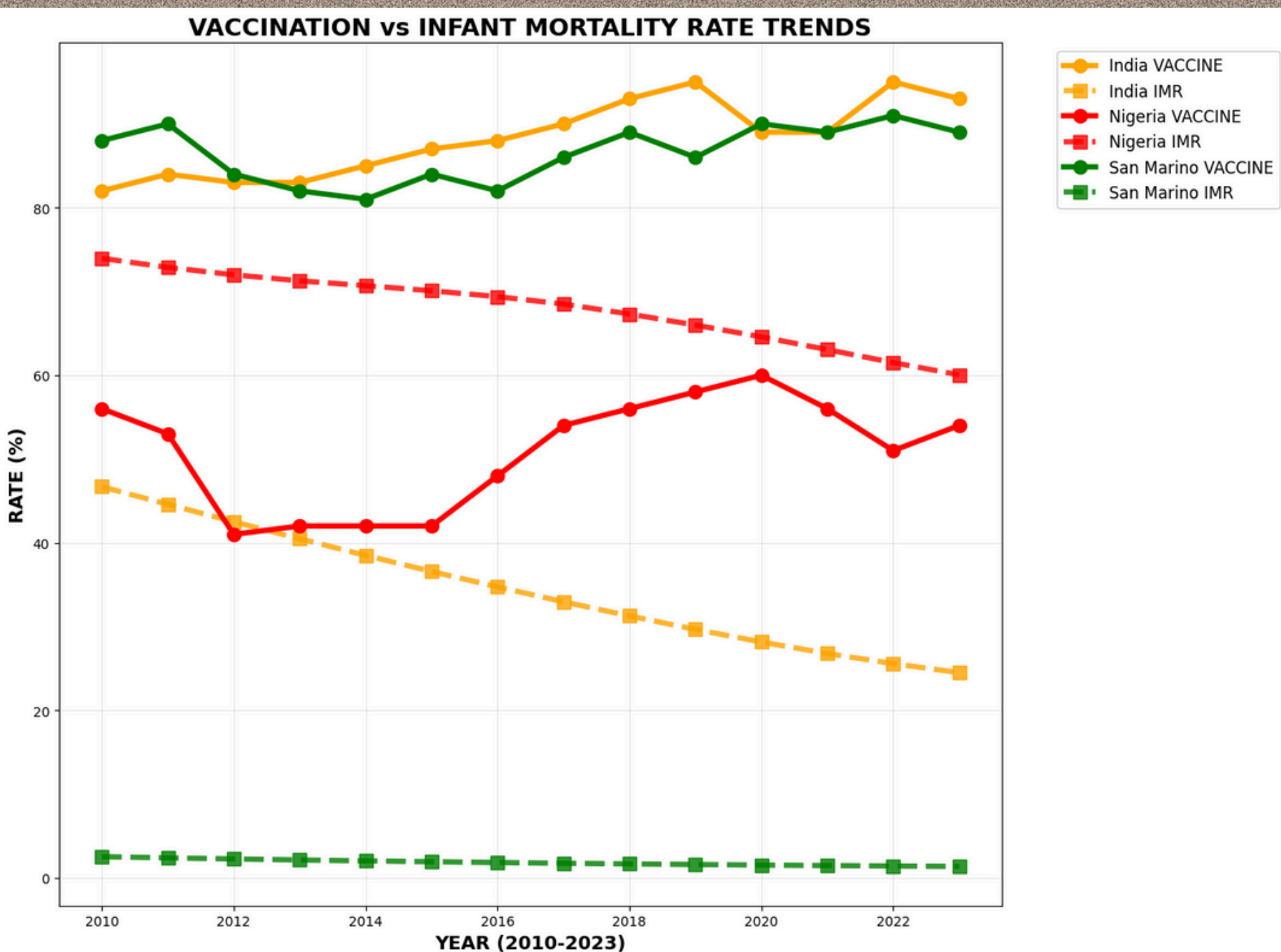
$P = 0.000 \rightarrow$  **STATISTICALLY SIGNIFICANT**  
(99.9% confidence - NOT random chance)

- Visualisation :** Reveals a strong negative correlation – higher measles vaccination coverage corresponds to lower infant mortality rates.
- Key Insight:** Vaccination programs play a critical role in reducing infant mortality worldwide.





# Specific Country Analysis: India, Nigeria, and San Marino



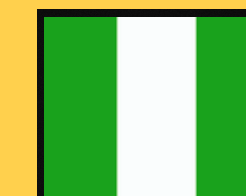
- **India shows substantial improvement:** ~90% vaccination coverage with infant mortality reduced to 25 per 1,000.

- Universal Immunization Programme
- Massive campaigns + healthcare infrastructure
- Economic growth → better nutrition/health



- **Nigeria faces challenges:** ~62% vaccination with higher infant mortality around 70 per 1,000.

- zero-dose children globally
- Conflict zones + poor access
- Weak health systems



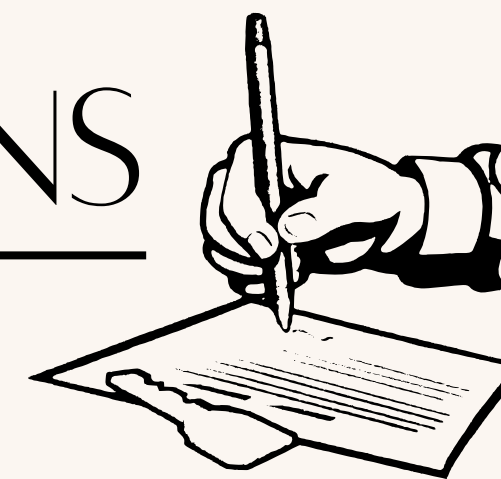
- **San Marino represents a global best:** ~98% vaccination with infant mortality around 1.5 per 1,000 (largely constant due to small population).

- High-income EU nation
- Universal healthcare + small population
- Perfect compliance





# CONCLUSIONS AND POLICY IMPLICATIONS



- **Analysis shows strong inverse relationship:**
  - **$r=-0.617$**  correlation between vaccination coverage and infant mortality across 2,435 country-years (2010-2023).
  - Every **10%** increase in measles vaccination saves **4-5** infant lives per 1,000 births, explaining **62%** of mortality differences
- **Africa faces 18X mortality gap:**
  - **(55 deaths/1k vs Europe's 3 deaths/1k)** due to **28%** coverage shortfall.
  - Globally, **14.3M** infants remain zero-dose and **20M** miss measles vaccine yearly
- **Vaccines drive 40% of infant survival gains:**
  - Worldwide, saving **59M** lives since 2000 at \$52 return per \$1 invested.
- **Policy Priority:**
  - Target Africa/South Asia with Gavi-funded campaigns (1.2B kids protected).
  - Data-driven immunization closes gaps fastest.



# NAKUL'S WORK



# Data Cleaning & Preprocessing

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
df = pd.read_csv("/content/drive/MyDrive/Life Expectancy Data.csv")
```

```
num_cols = df.select_dtypes(include="number").columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())

cat_cols = df.select_dtypes(include='object').columns
df[cat_cols] = df[cat_cols].fillna(df[cat_cols].mode().iloc[0])
```

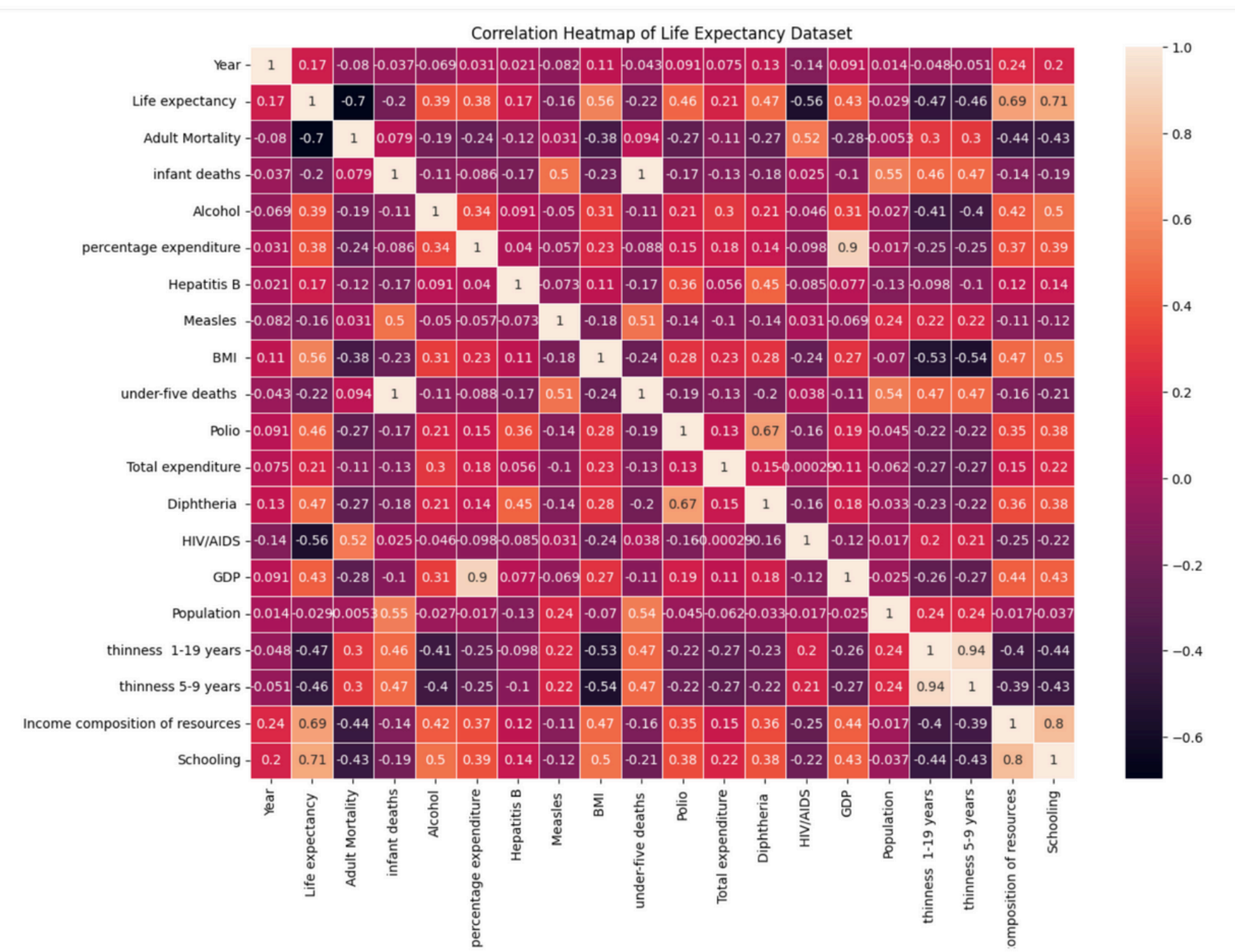
```
df.drop_duplicates(inplace=True)
```

- Checked for missing values to assess data quality.
- Filled missing numerical values with the median to avoid outlier.
- Filled missing categorical values with the mode for consistency.
- Removed duplicate records to maintain data integrity.
- Prepared a clean and reliable dataset for further EDA and modeling.

```
df.isnull().sum()
```

|                                 | 0   |
|---------------------------------|-----|
| Country                         | 0   |
| Year                            | 0   |
| Status                          | 0   |
| Life expectancy                 | 10  |
| Adult Mortality                 | 10  |
| infant deaths                   | 0   |
| Alcohol                         | 194 |
| percentage expenditure          | 0   |
| Hepatitis B                     | 553 |
| Measles                         | 0   |
| BMI                             | 34  |
| under-five deaths               | 0   |
| Polio                           | 19  |
| Total expenditure               | 226 |
| Diphtheria                      | 19  |
| HIV/AIDS                        | 0   |
| GDP                             | 448 |
| Population                      | 652 |
| thinness 1-19 years             | 34  |
| thinness 5-9 years              | 34  |
| Income composition of resources | 167 |
| Schooling                       | 163 |

# Correlation Heatmap of Life Expectancy Dataset

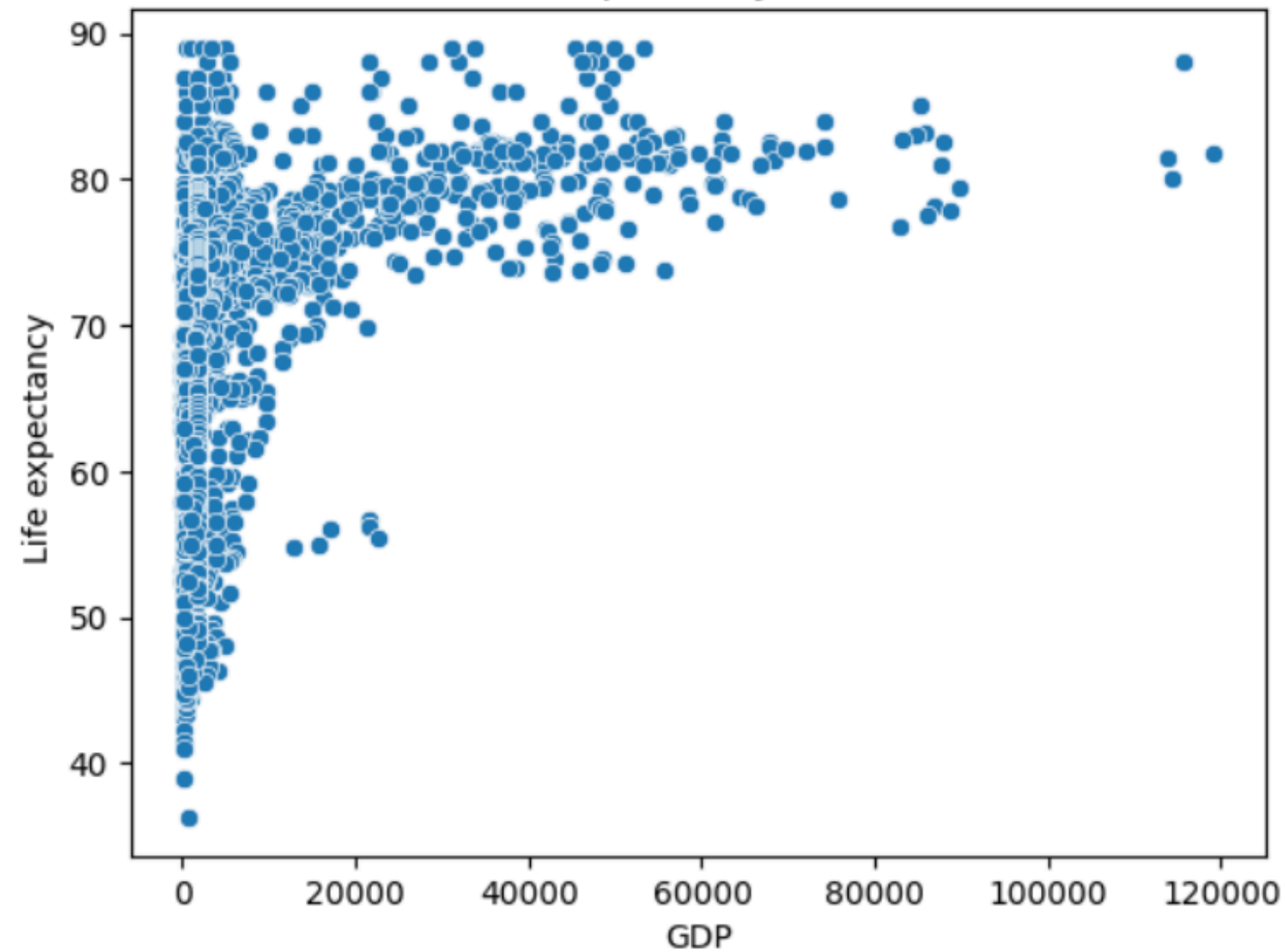


- Calculates the correlation between all numerical variables.
- Visualizes these correlations as a heatmap to show how strongly features are related to each other.
- Helps identify which factors have strong positive or negative relationships with life expectancy.

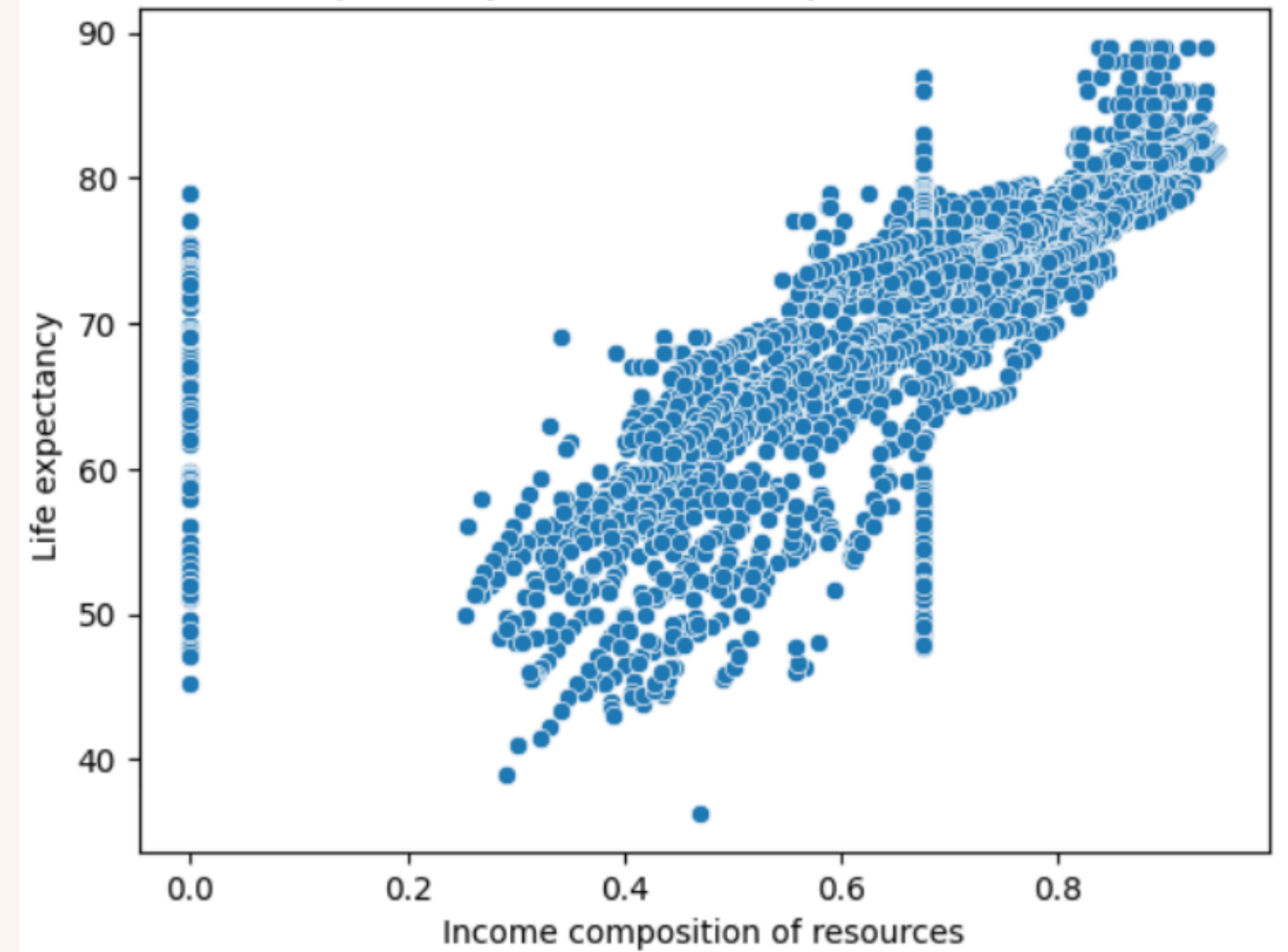
| Correlation Value Interpretation |                               |
|----------------------------------|-------------------------------|
| Correlation Value                | Meaning                       |
| +1.0                             | Perfect positive relationship |
| +0.5 to +1.0                     | Strong positive correlation   |
| +0.2 to +0.5                     | Moderate positive correlation |
| 0.0                              | No linear relationship        |
| -0.2 to -0.5                     | Moderate negative correlation |
| -0.5 to -1.0                     | Strong negative correlation   |
| -1.0                             | Perfect negative relationship |

# Scatter plots:

Life Expectancy vs GDP



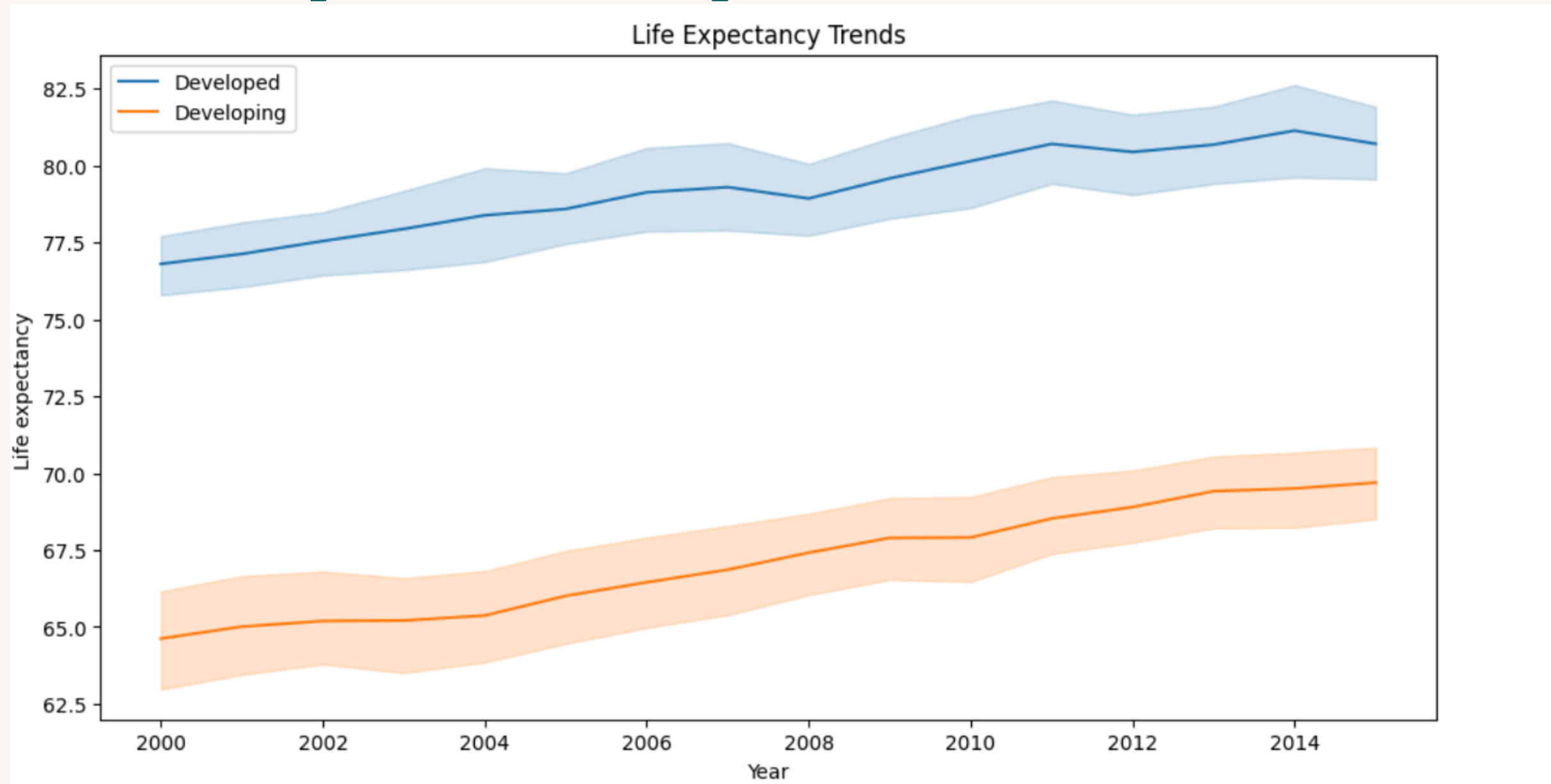
Life Expectancy vs Income composition of resources



## Life Expectancy vs Income Composition & GDP (Key Insights)

- Higher income composition is strongly linked to higher life expectancy.
- Better access to resources leads to healthier and longer lives.
- Countries with higher GDP generally show higher life expectancy.
- After a certain GDP level, increases do not significantly raise life expectancy (saturation effect).
- Low-GDP countries consistently cluster with lower life expectancy.

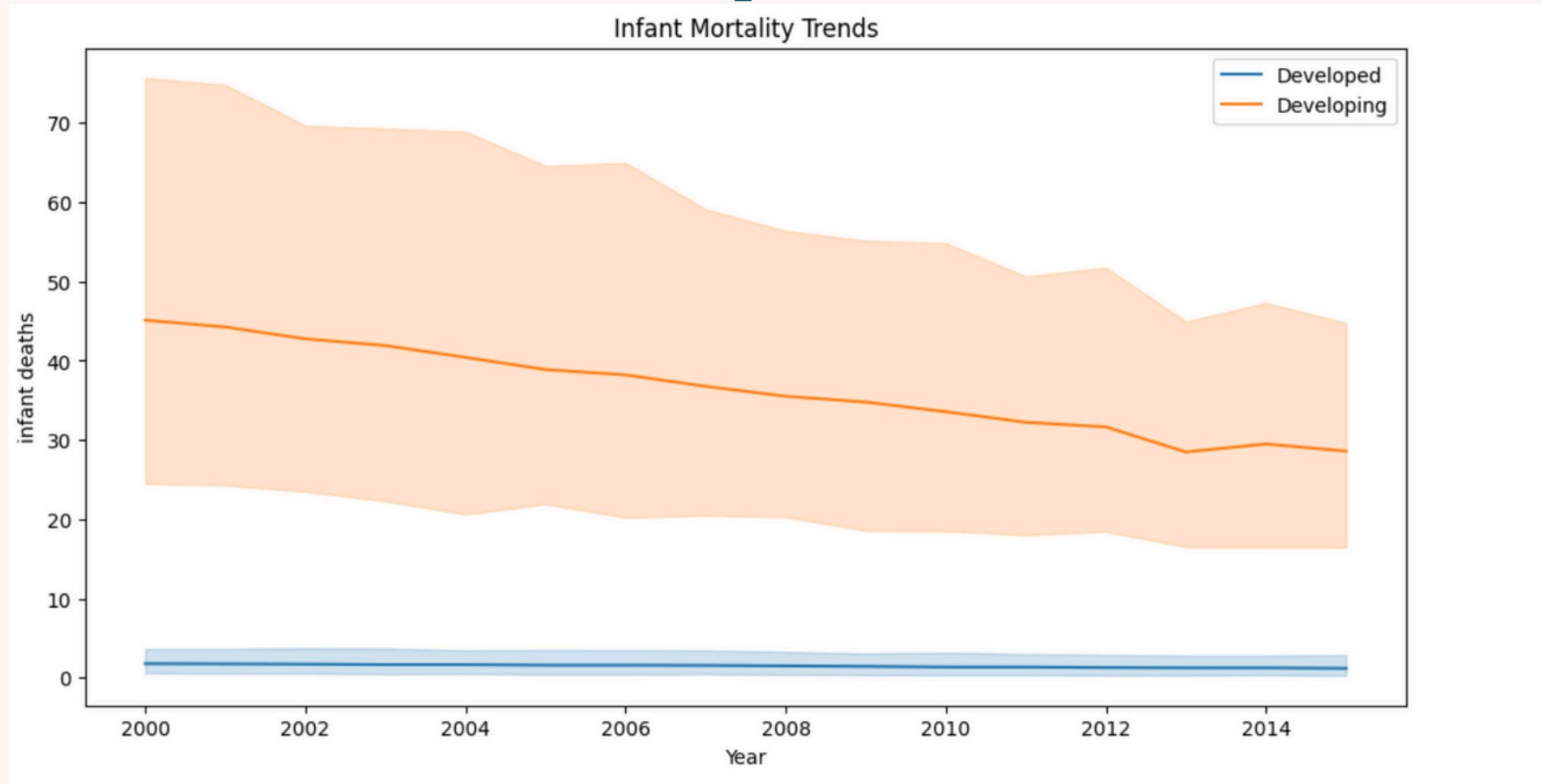
# "Life Expectancy Trends":



## Life Expectancy Trends (Developed vs Developing)

- Life expectancy rises steadily for both groups from 2000–2015.
- Developed countries remain consistently higher ( $\approx 78$ – $82$ ).
- Developing countries start lower ( $\approx 65$ ) but show clear improvement.
- The gap stays, but both groups move upward over time.

# "Infant Mortality Trends"



## Infant Mortality Trends (2000-2015)

- Developed: Rate is consistently very low (around 4-5) and stable.
- Developing: Rate is much higher (starting ~ 45), but shows a steady, significant decline (ending < 30).
- Conclusion: The gap is huge but is slowly narrowing due to improvements in developing nations.

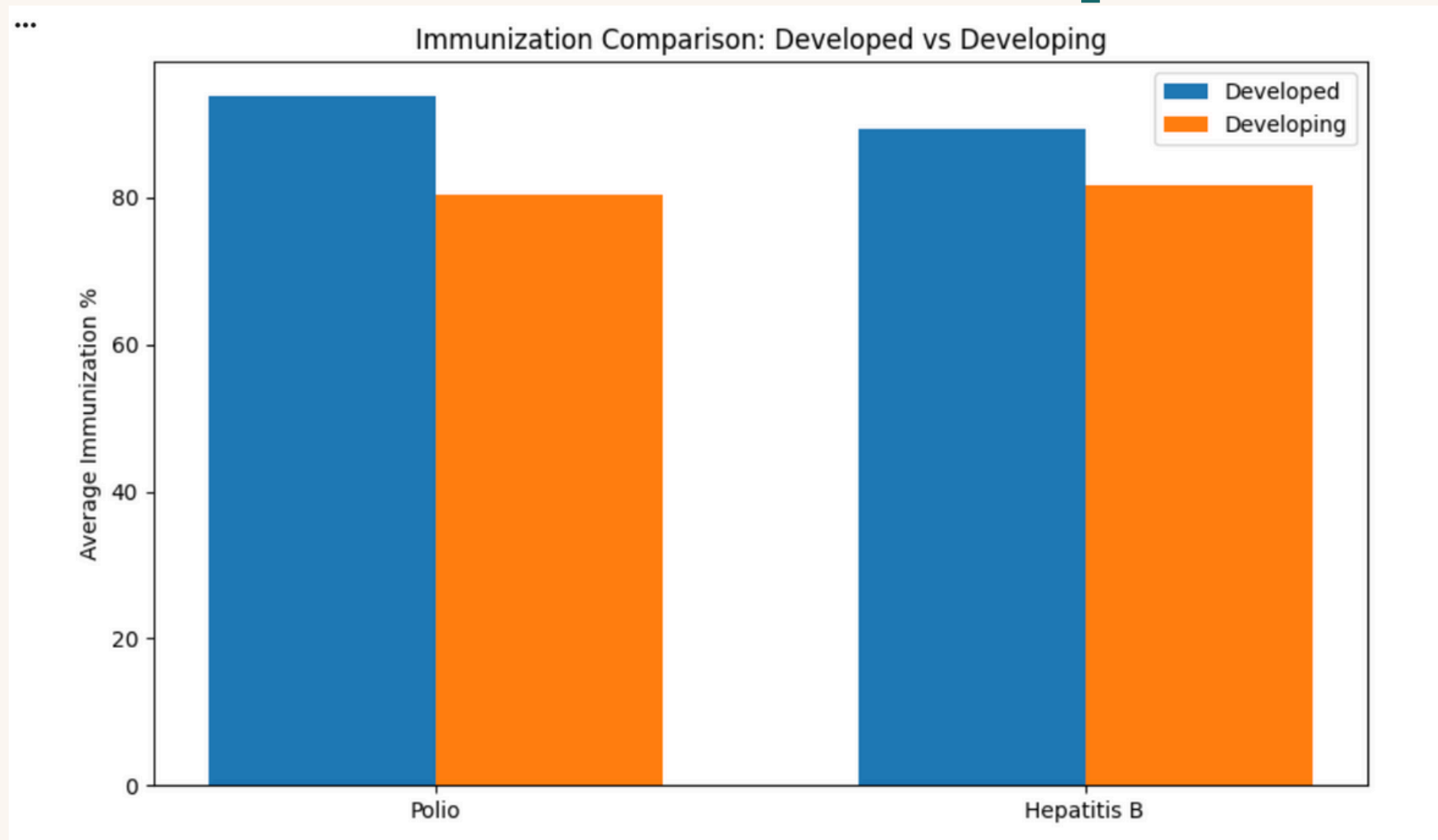
# Life Expectancy GAP (Developed - Developing):



## Life Expectancy Gap (2000–2015)

- The gap peaked around 2004 at roughly 13 years.
- From 2004–2008, it declined sharply to about 11.5 years.
- By 2014–2015, the gap narrowed further to around 11 years.
- Overall, the gap in 2015 was smaller than in 2000, showing gradual convergence.

# Immunization Comparison: Developed vs Developing



## Immunization Comparison: Developed vs Developing (Polio & Hepatitis B)

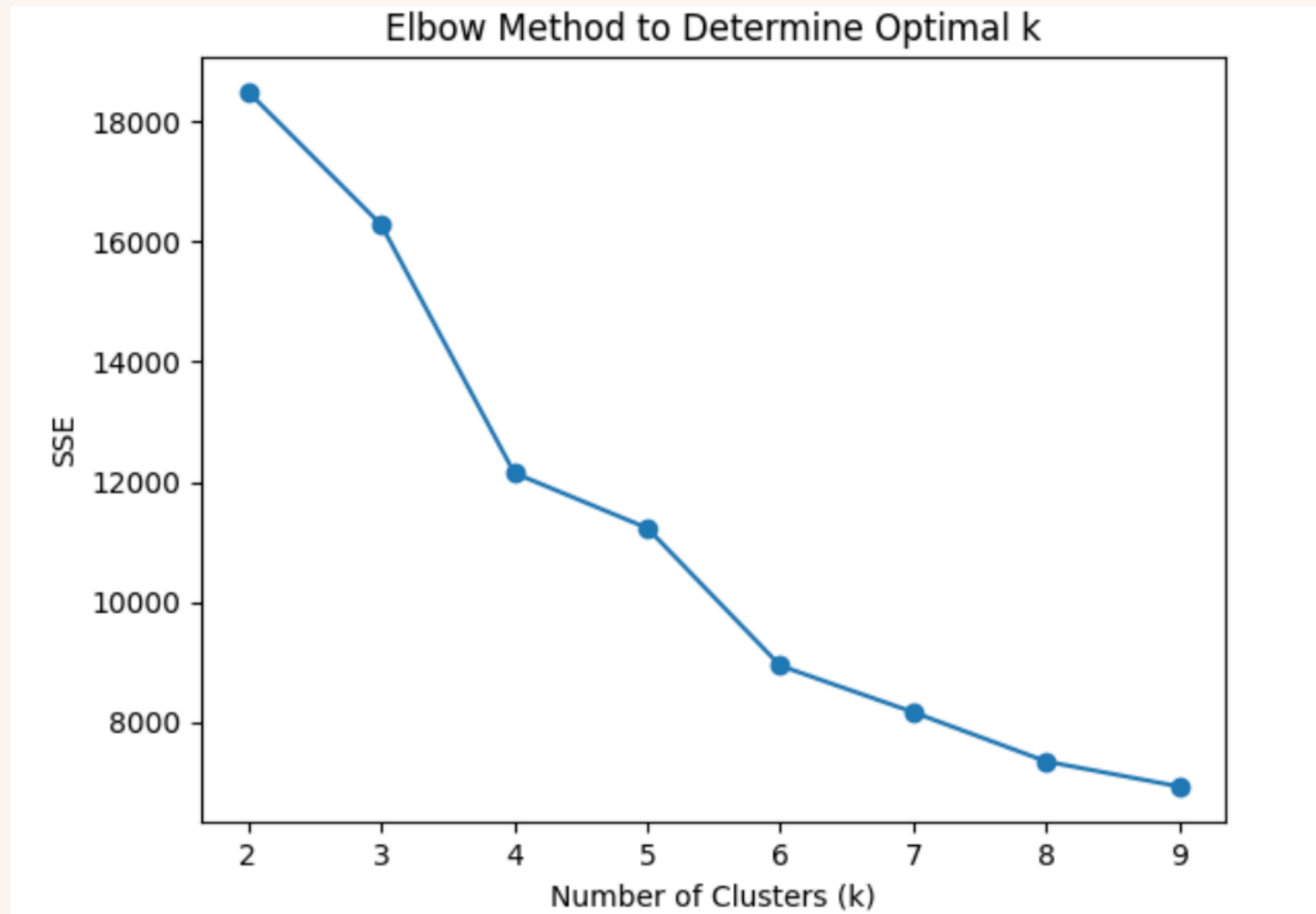
- Developed countries show higher immunization rates for both Polio and Hepatitis B, consistently above 88–90%.
- Developing countries have lower coverage, around 80–82%, showing a noticeable gap.
- The difference indicates stronger healthcare systems and vaccination programs in developed nations.
- Despite the gap, developing countries still maintain relatively high coverage, reflecting ongoing global immunization efforts.

# Country Development Classification Using K-Means Clustering (with PCA Visualization)

```
▶ from sklearn.preprocessing import StandardScaler
from sklearn.cluster import KMeans
features = [
    'Life expectancy ',
    'Adult Mortality',
    'infant deaths',
    'Income composition of resources',
    'Schooling',
    'percentage expenditure',
    ' HIV/AIDS',
    ' thinness 5-9 years',
    'GDP'
]
data = df[features].dropna()
scaler = StandardScaler()
scaled_data = scaler.fit_transform(data)
```

- Selects important numeric features from the dataset (life expectancy, mortality, schooling, GDP, etc.)
- Removes rows with missing values to keep the data clean (dropna).
- Creates a StandardScaler to normalize all features.
- Standardizes the data so every feature has equal weight during K-Means clustering.

# Elbow Method to Determine Optimal k:



- Creates an empty list sse to store the Sum of Squared Errors for each k value.
- Defines a range of possible cluster numbers (k = 2 to 9).
- Loops through each k, fits a K-Means model, and records its SSE (inertia\_).
- Plots k vs SSE to form the Elbow Curve.
- Uses the elbow point in the graph to determine the optimal number of clusters.

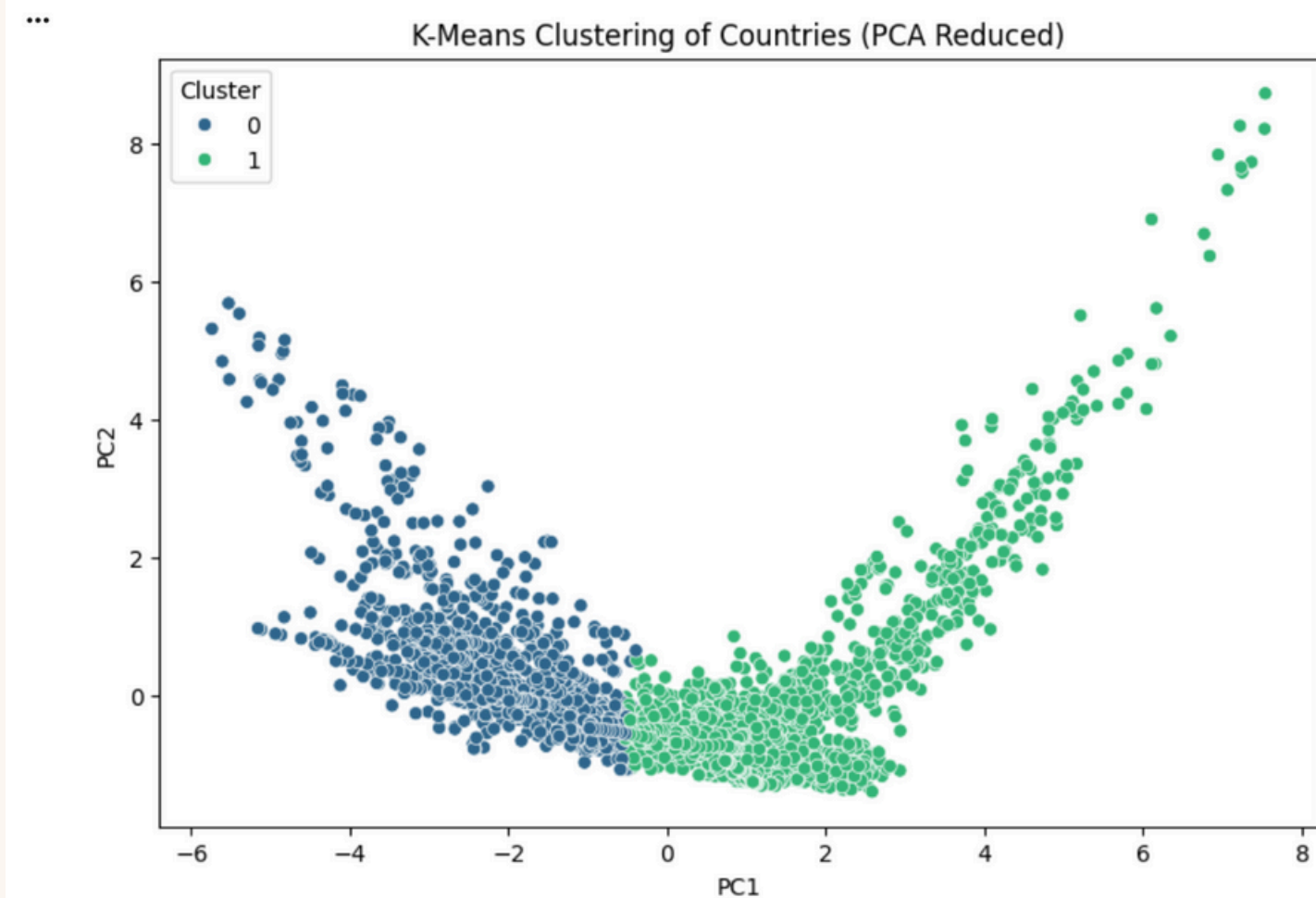
# K-Means Clustering of Countries

```
from sklearn.decomposition import PCA
```

```
pca = PCA(n_components=2)  
reduced = pca.fit_transform(scaled_data)
```

```
df['PC1'] = reduced[:, 0]  
df['PC2'] = reduced[:, 1]
```

```
plt.figure(figsize=(9, 6))  
sns.scatterplot(data=df, x='PC1', y='PC2', hue='Cluster', palette='viridis')  
plt.title("K-Means Clustering of Countries (PCA Reduced)")  
plt.show()
```



- Imports PCA to reduce high-dimensional data into fewer components.
- Creates a PCA model that keeps 2 components (PC1 and PC2).
- Applies PCA on the scaled data to convert it into 2-dimensional form
- Adds PC1 and PC2 values back into the dataframe for visualization
- Plots a scatter graph using PC1 and PC2, coloring points by their cluster labels
- Helps visualize how countries are grouped by K-Means in a 2D space.

# Year-wise Cluster Analysis for Developing Countries

```
▶ developing = df[df['Status'] == 'Developing']  
cluster_by_year = developing.groupby(['Year'])['Cluster'].value_counts().unstack()  
print(cluster_by_year)
```

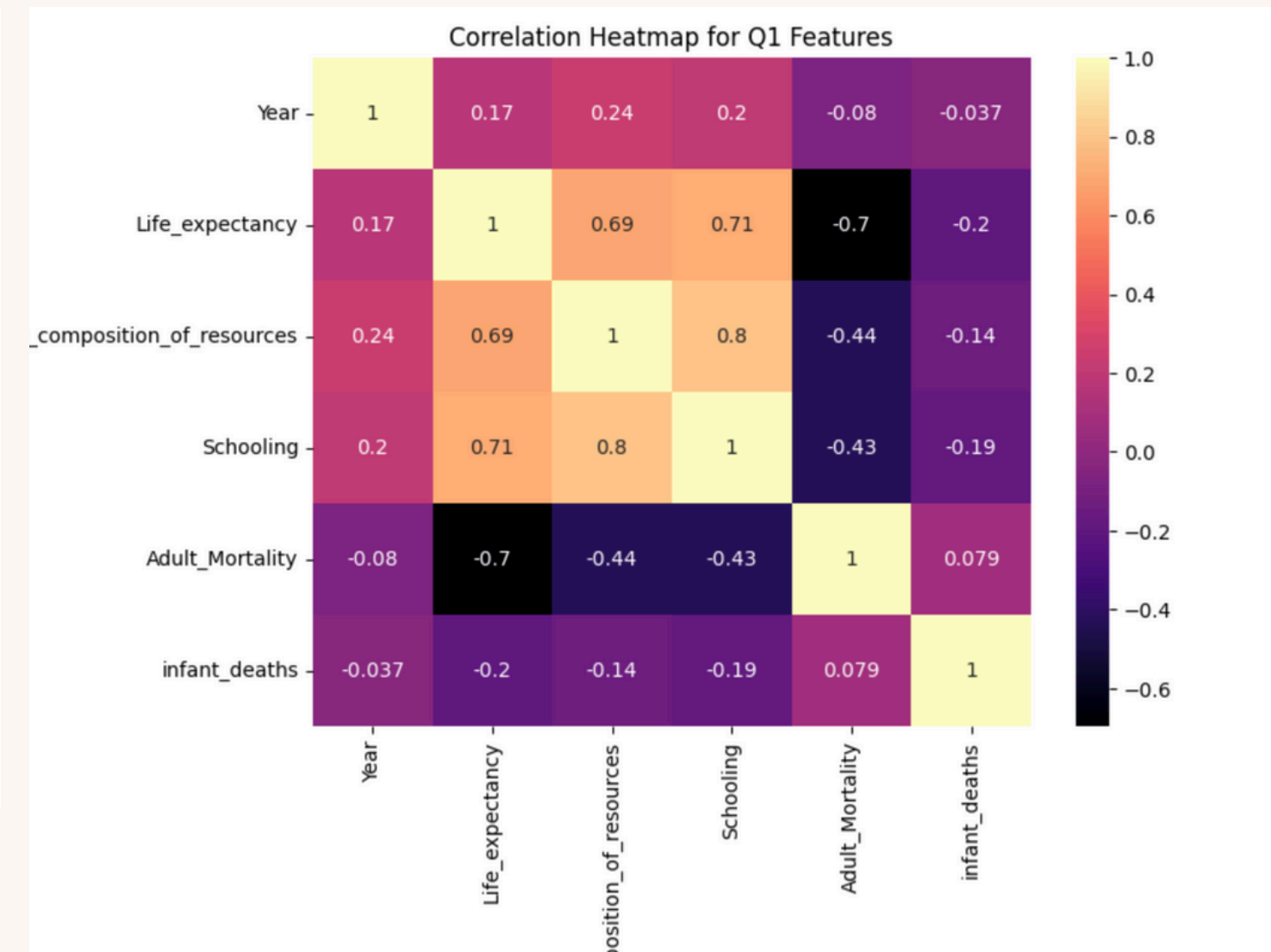
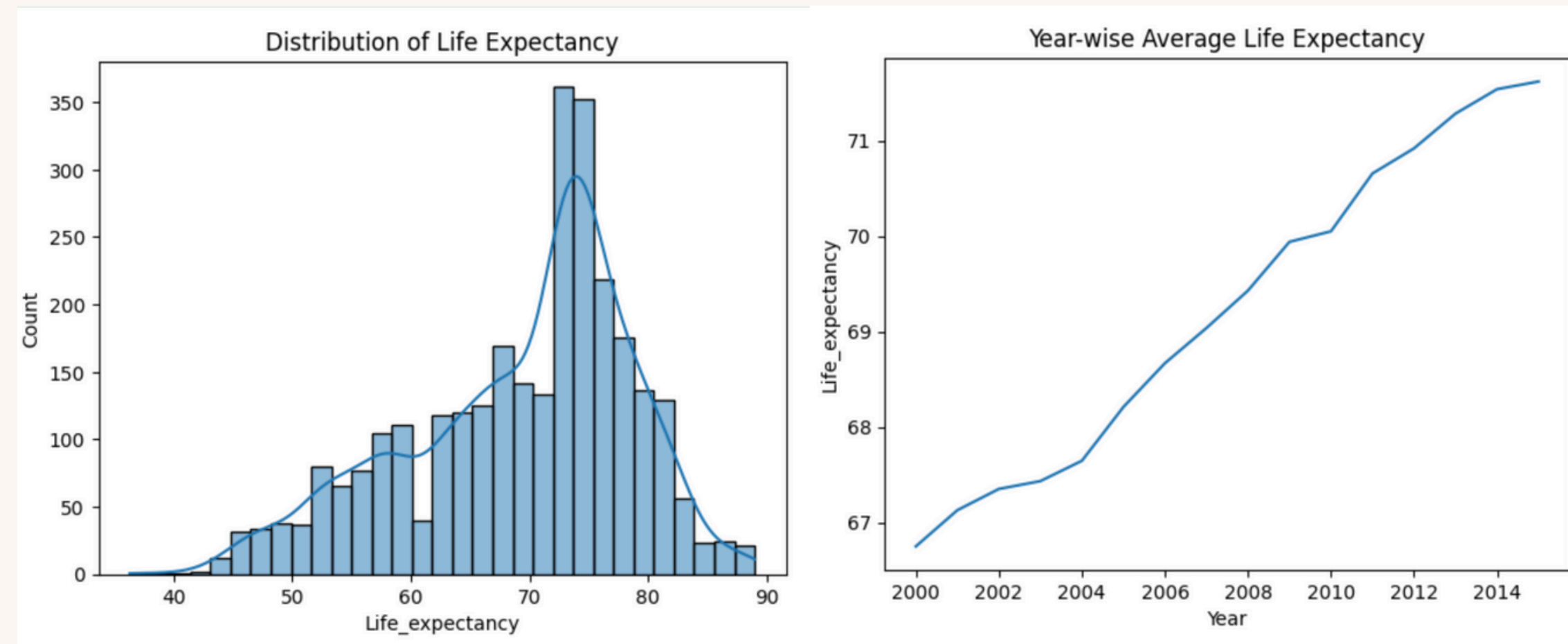
```
... Cluster    0    1  
Year  
2000         92   59  
2001         78   73  
2002         76   75  
2003         78   73  
2004         74   77  
2005         68   83  
2006         65   86  
2007         65   86  
2008         65   86  
2009         62   89  
2010         63   88  
2011         62   89  
2012         59   92  
2013         63   98  
2014         58   93  
2015         57   94
```

- Filters the dataframe to keep only rows where Status is "Developing".
- Groups these developing countries by Year.
- Counts how many countries fall into each Cluster for every year.
- Uses unstack() to convert the counts into a clean table format (years as rows, clusters as columns).
- Prints the table to show how developing countries shift between clusters over time.

# ANUJ'S WORK

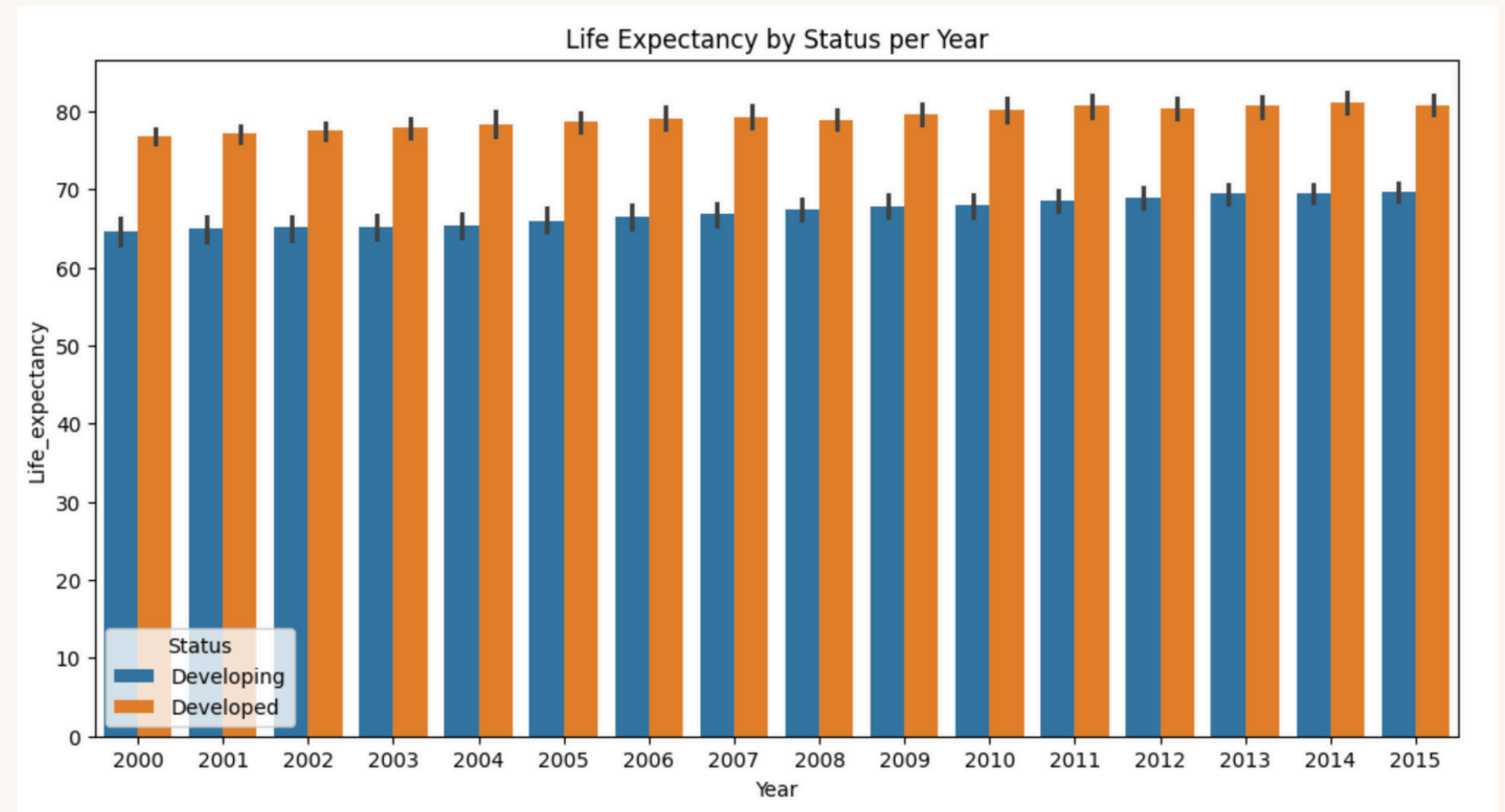
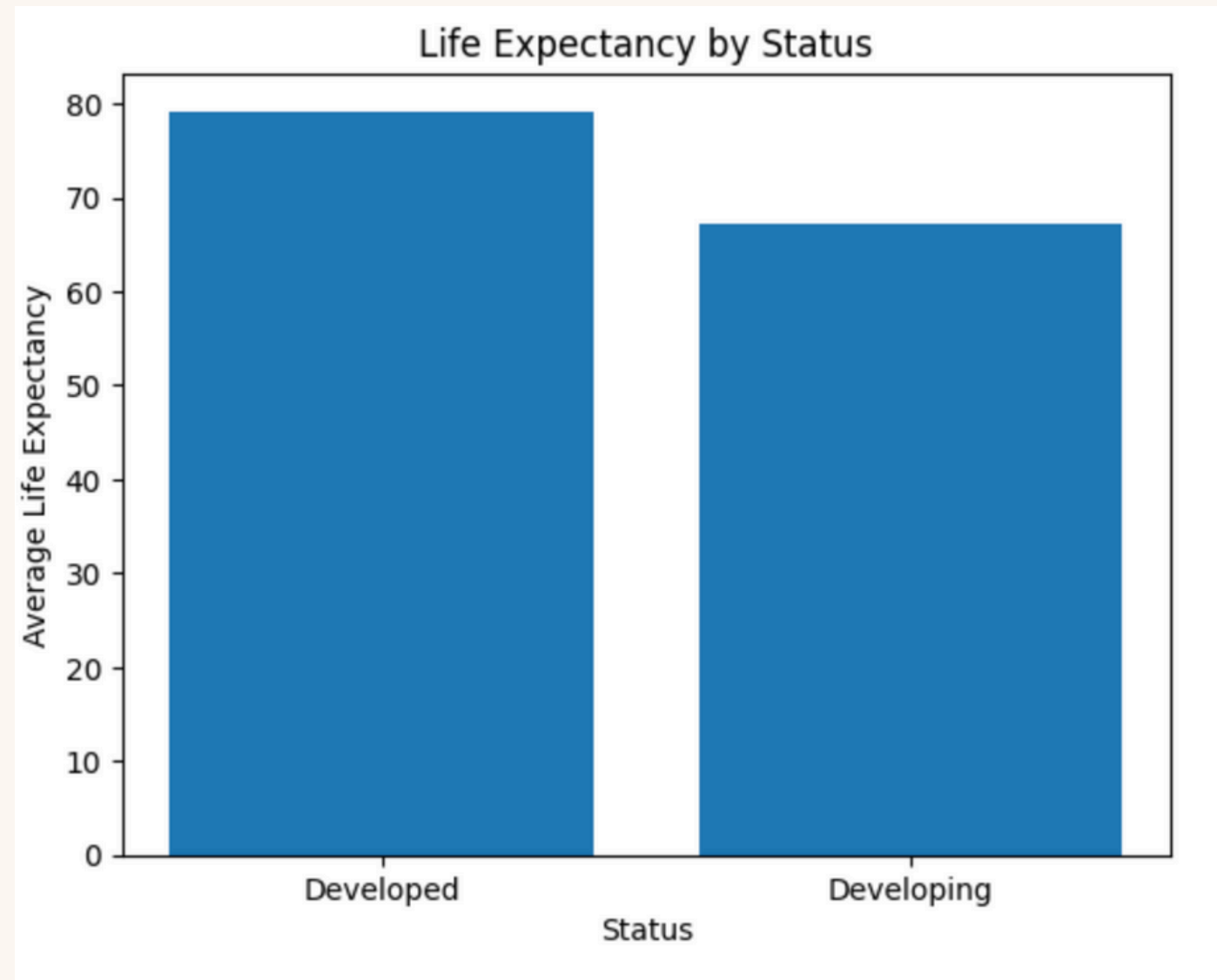


# Insights From Life Expectancy Dataset



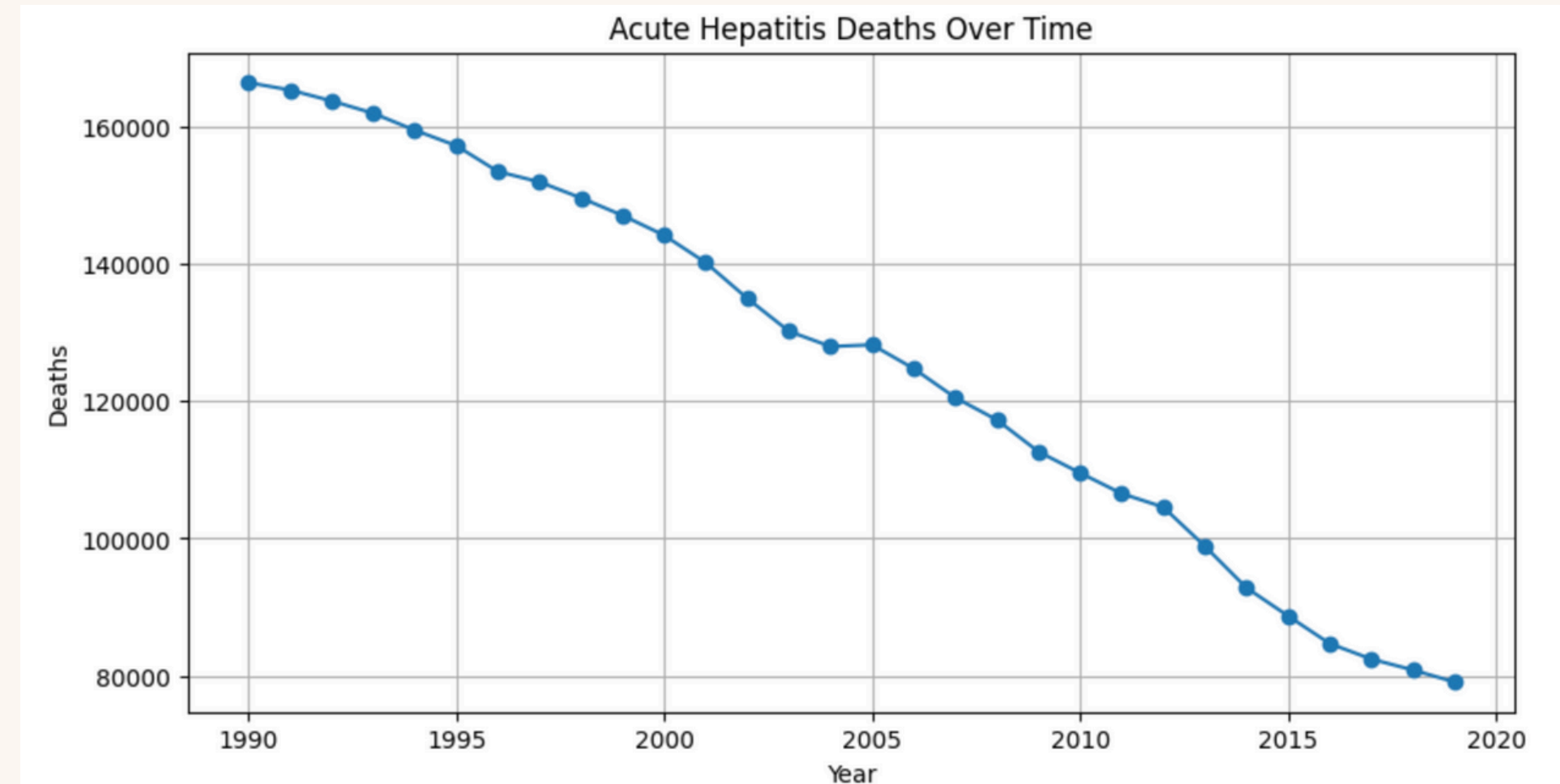
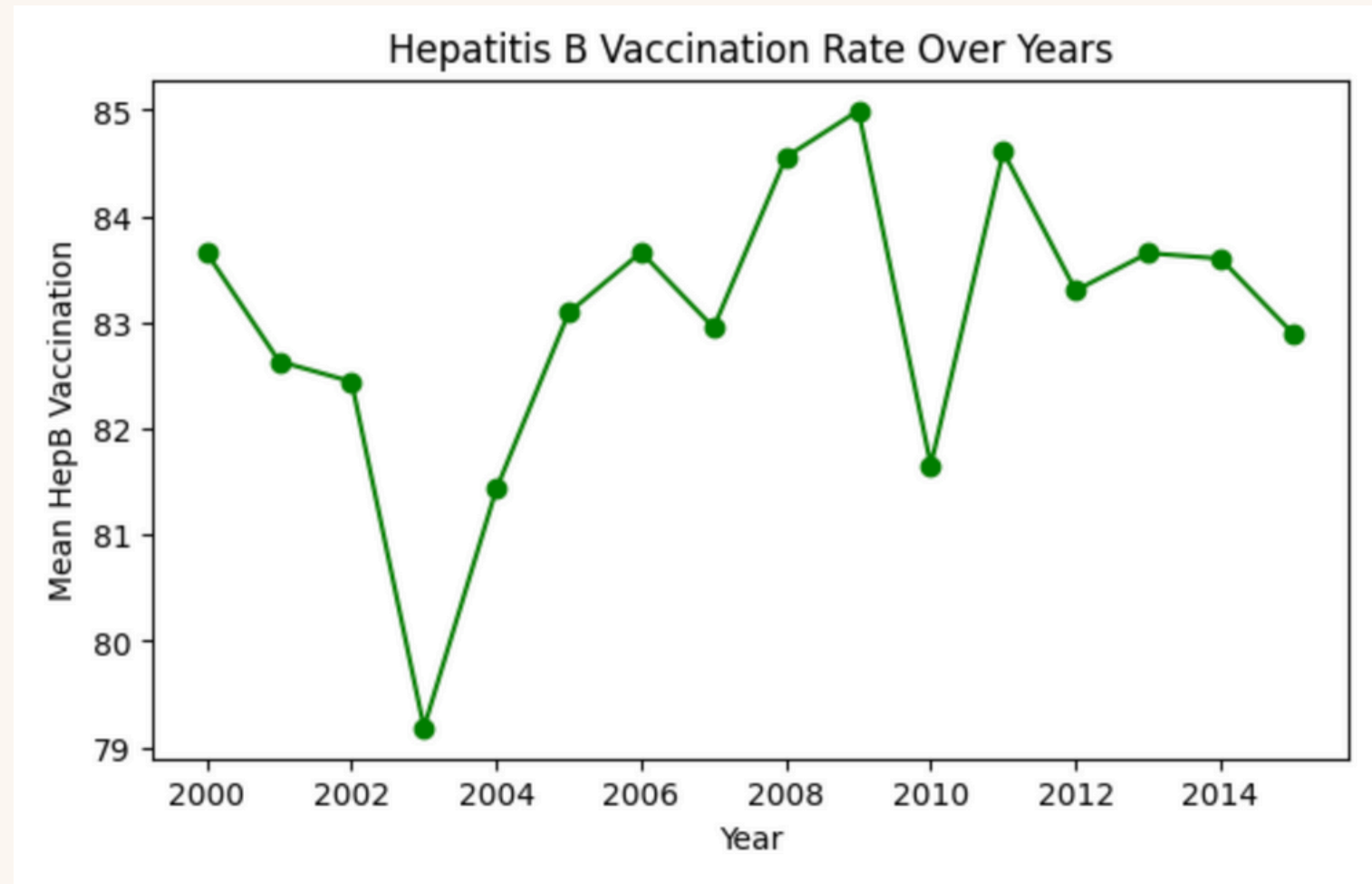
- **Correlation Heatmap:** Better healthcare and socio-economic factors show strong positive correlation with life expectancy, while mortality-related factors correlate negatively.
- **Distribution Plot:** Most countries fall in the 60–75 year range, with fewer nations achieving 80+ years.
- **Year-wise Trend:** Life expectancy shows a consistent upward trend over the years globally.

# Insights From Life Expectancy Dataset



- **Developed vs Developing over Years:** Developed countries consistently show higher life expectancy across all years, with a gap of around 10–15 years, though both categories show gradual improvement over time.
- **Trend Stability:** Developed nations have a more stable and slowly increasing trend, while developing nations show slightly more variation year-to-year but still a clear upward growth.
- **Overall Average Comparison:** On average, developed countries have significantly higher life expectancy (~79 years) compared to developing countries (~72 years), highlighting disparities in healthcare, infrastructure, and living conditions.

# Impact of Hepatitis B Vaccination on Mortality Trends



- Hepatitis B vaccination coverage remains consistently high, mostly above 82%, indicating strong global immunization efforts.
- Acute Hepatitis deaths show a steady and significant decline from 1990 to 2020, dropping from ~168,000 to ~79,000.
- **The downward trend in deaths aligns with sustained vaccination levels, suggesting that widespread Hepatitis B immunization has played a key role in reducing hepatitis-related mortality over time.**

# Top Causes of Death



- **Cardiovascular diseases are the leading global killer.**
- **Chronic diseases like cancer and chronic respiratory diseases also contribute heavily.**
- **Infectious diseases remain important but no longer dominate worldwide mortality.**

# Global Shifts in Causes of Death Over Time

Top 10 diseases with the largest reduction over time:

1. diarrheal\_diseases: -1361263.0
2. neonatal\_disorders: -1122709.0
3. lower\_respiratory\_infections: -826824.0
4. tuberculosis: -597425.0
5. nutritional\_deficiencies: -505454.0
6. protein\_energy\_malnutrition: -443958.0
7. drowning: -223302.0
8. malaria: -197095.0
9. meningitis: -196125.0
10. maternal\_disorders: -106098.0

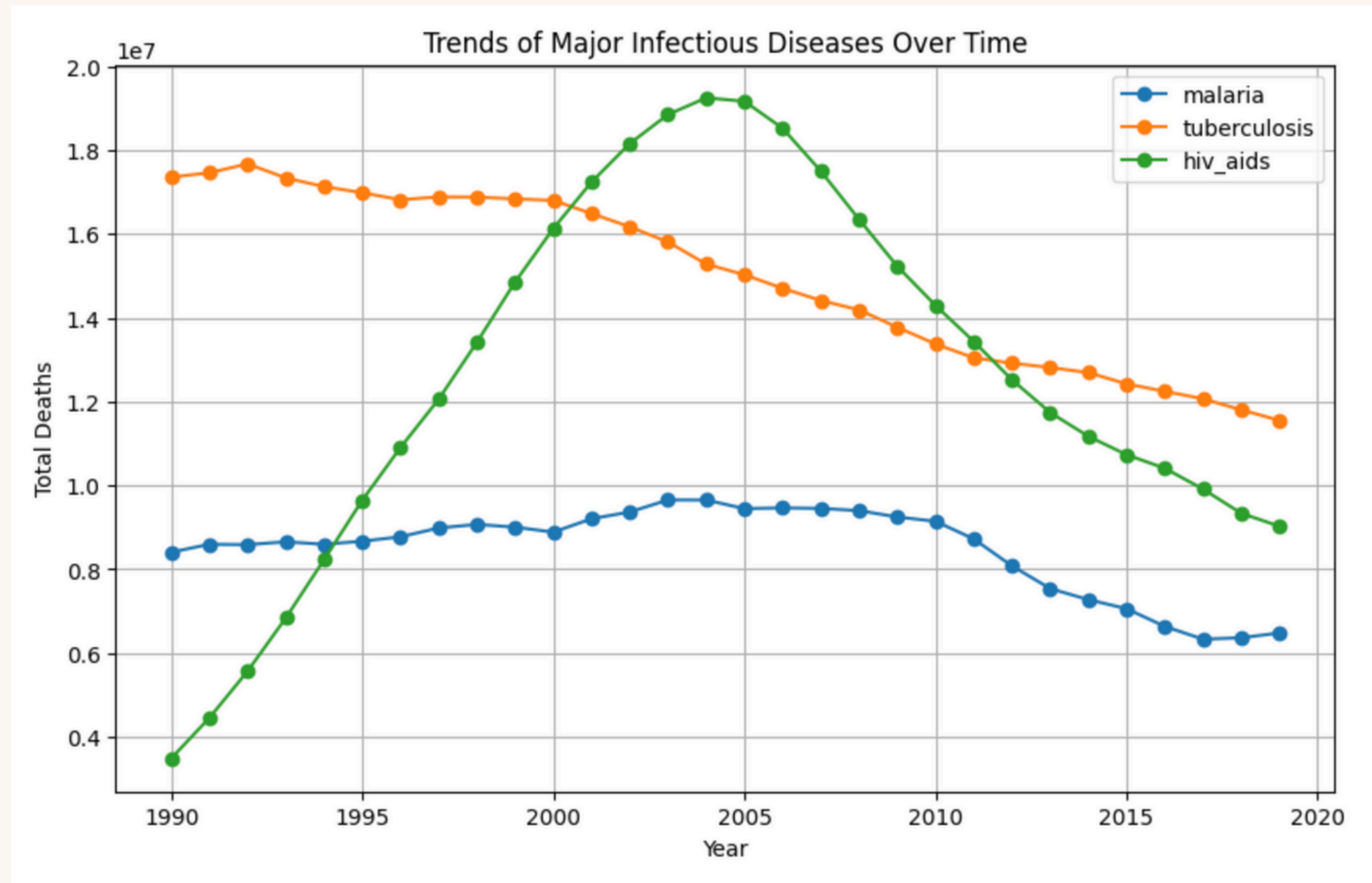
- The diseases with the largest reduction in deaths include:
- HIV/AIDS
- Malaria
- Tuberculosis
- Lower respiratory infections
- Strong evidence of successful global public health initiatives.

Top 10 diseases with the largest increase over time:

1. cardiovascular\_diseases: 6488634.0
2. neoplasms: 4318802.0
3. alzheimers\_disease\_and\_other\_dementias: 1060358.0
4. diabetes\_mellitus: 886724.0
5. chronic\_respiratory\_diseases: 878992.0
6. chronic\_kidney\_disease: 824168.0
7. digestive\_diseases: 701675.0
8. hiv\_aids: 527710.0
9. cirrhosis\_and\_other\_chronic\_liver\_diseases: 458746.0
10. parkinsons\_disease: 215333.0

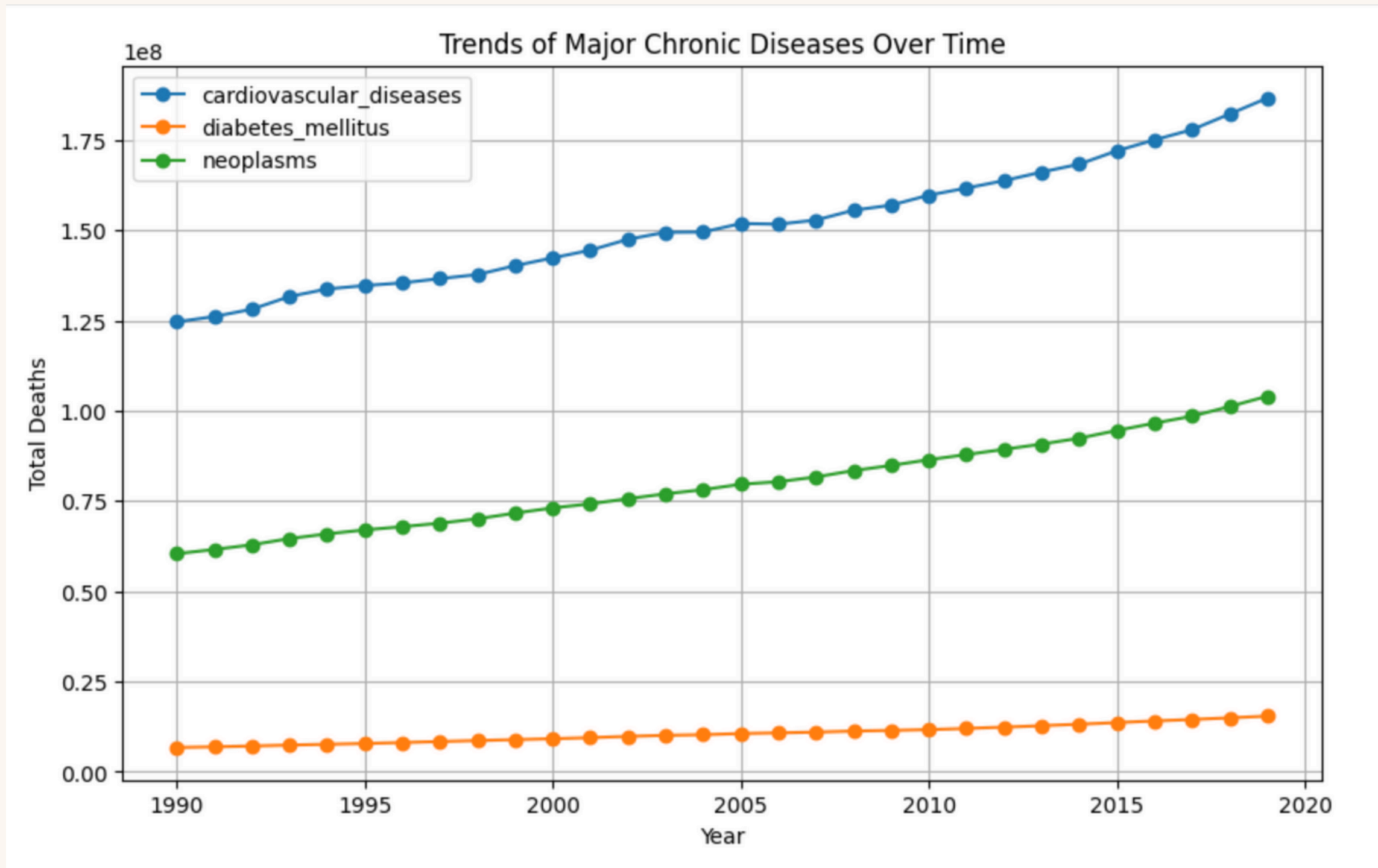
- The diseases with the largest increase in deaths include:
- Cardiovascular diseases
- Cancers (Neoplasms)
- Diabetes
- Chronic kidney disease
- These align with global lifestyle and aging patterns.

# Major Infectious Diseases Trends Over Time



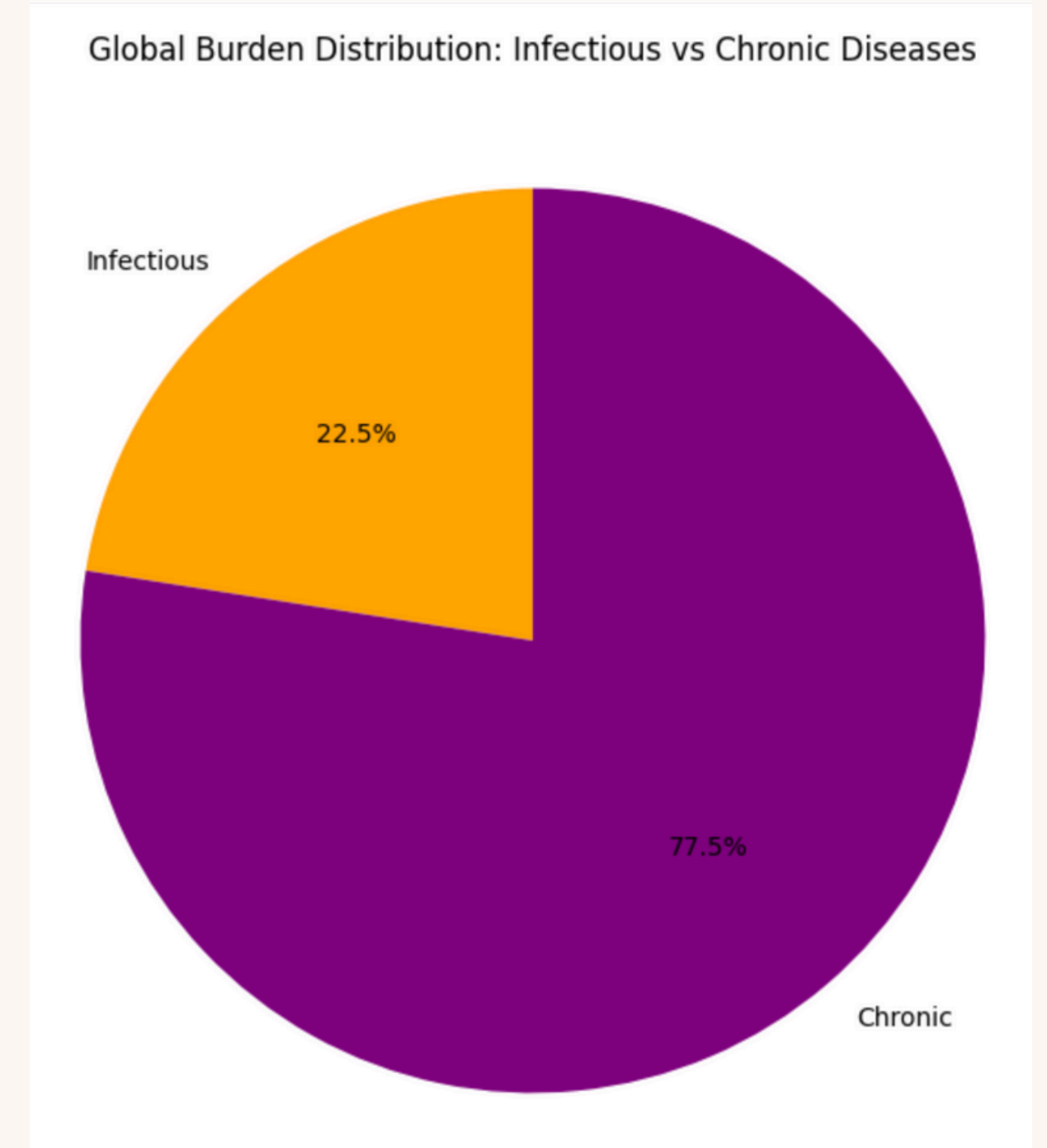
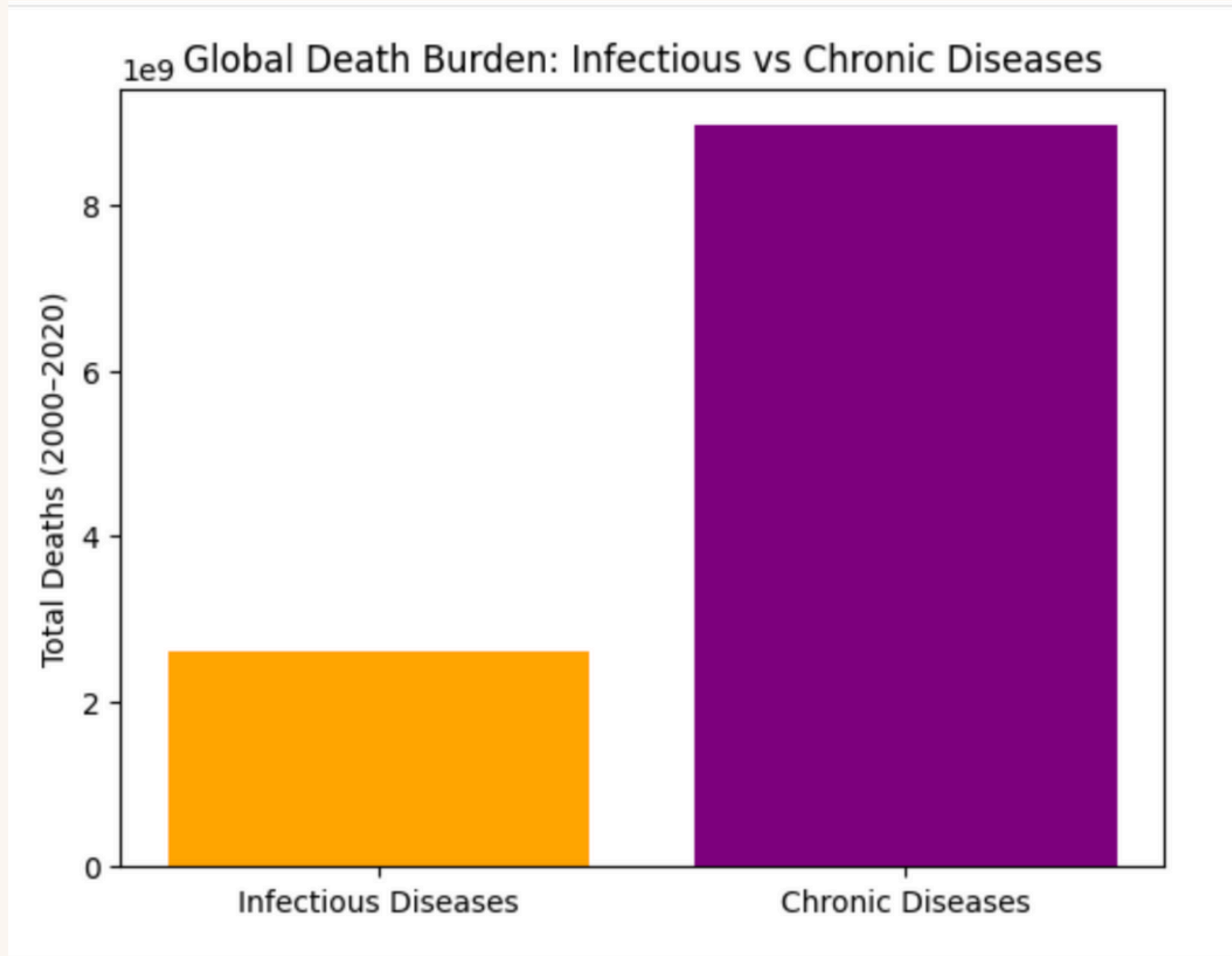
- All major infectious diseases show declining mortality.
- HIV deaths peaked around mid-2000s and then declined due to treatment access.
- Malaria and TB deaths have gradually reduced due to preventive and treatment programs.
- This indicates global progress in fighting infectious diseases.

# Major Chronic Diseases Trends Over Time



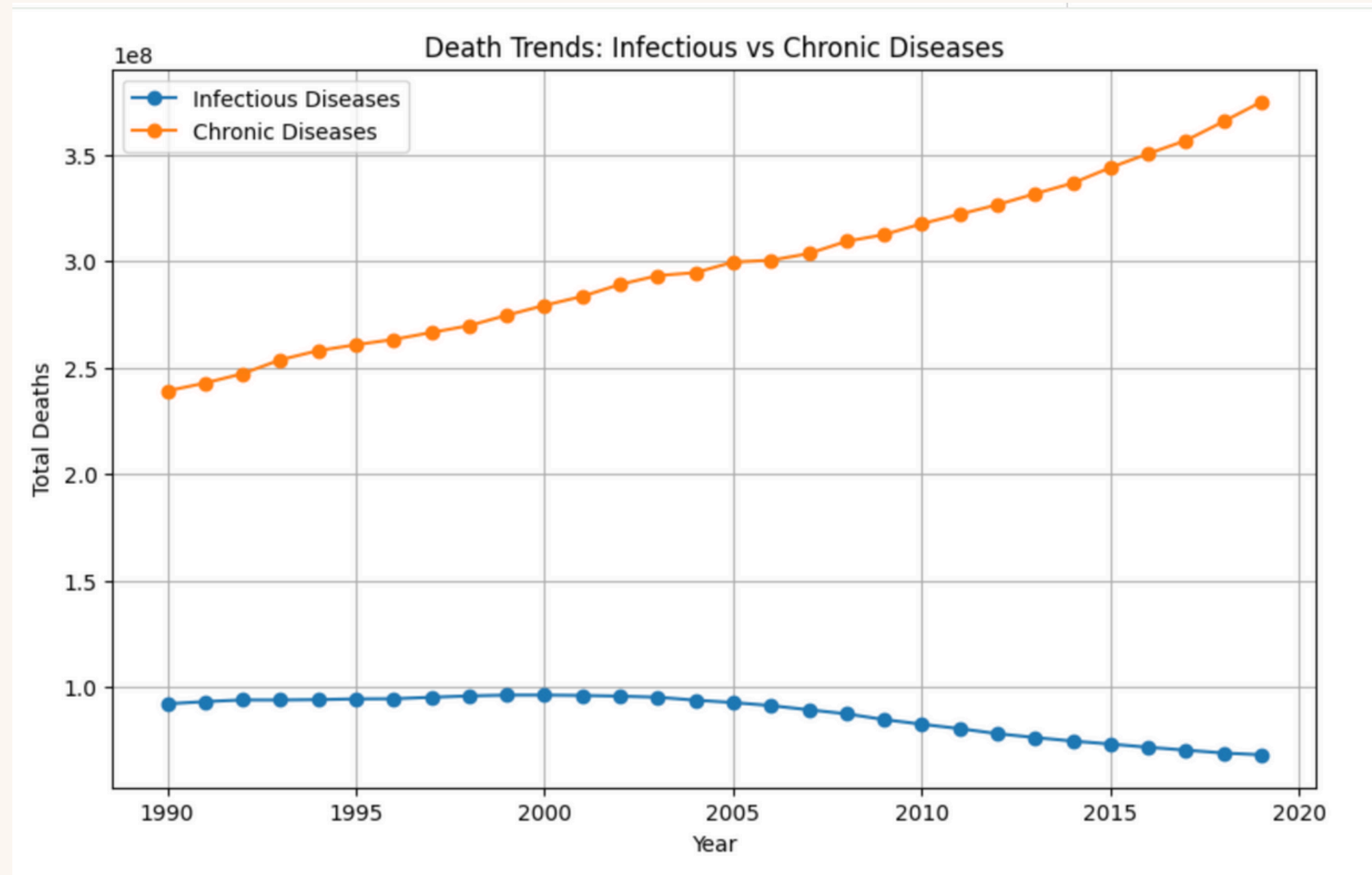
- Chronic diseases are increasing steadily worldwide.
- Heart disease remains the top global cause of death.
- Diabetes shows rapid increase — linked to lifestyle changes and aging.

# Infectious vs Chronic Comparison



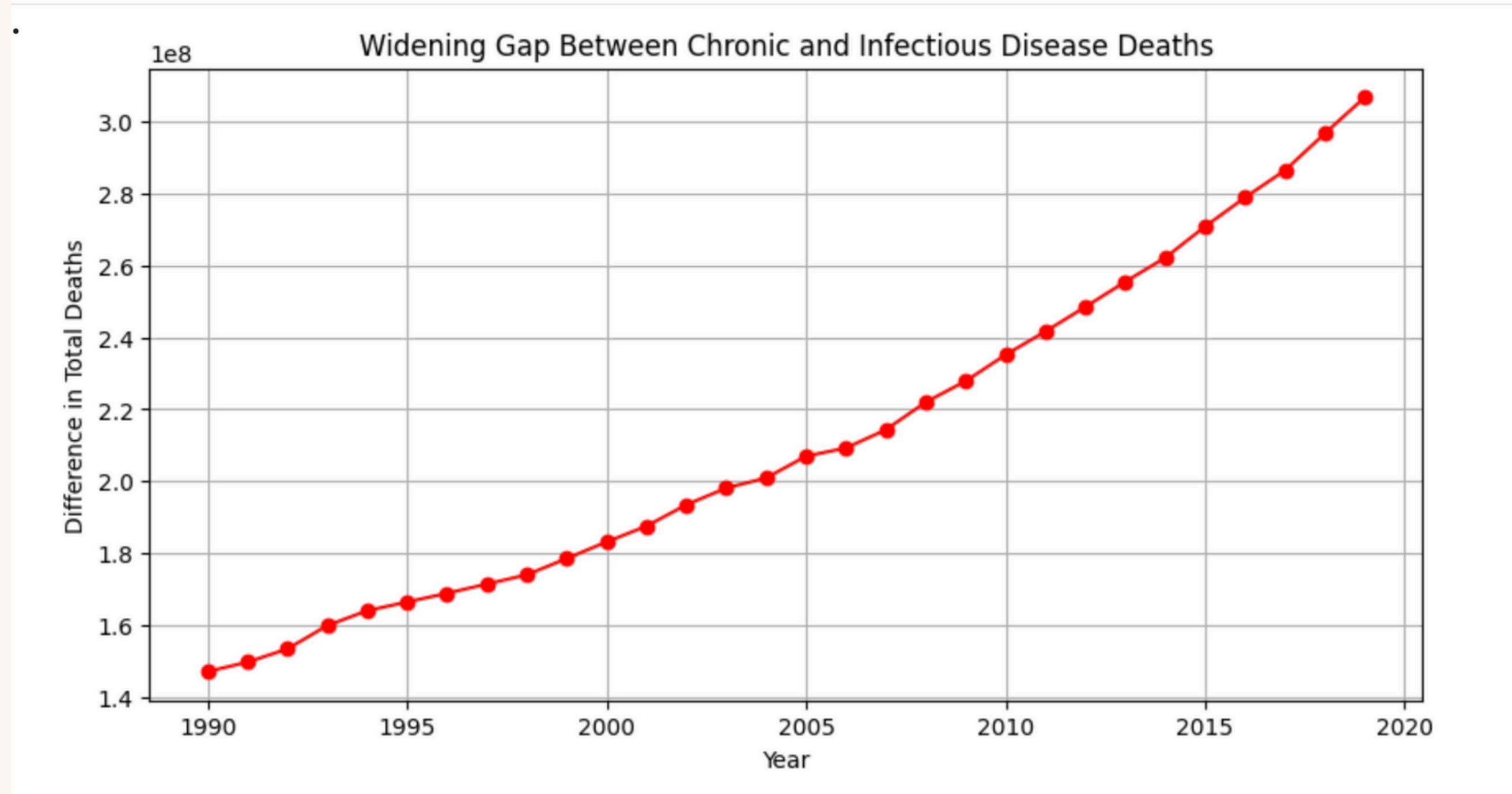
- Chronic diseases cause more deaths globally than infectious diseases.
- Infectious diseases remain deadly but have been declining over the years.
- This reflects the epidemiological transition where societies move from infectious → chronic disease burden.

# Global Health Is Shifting from Infectious → Chronic Diseases



- Infectious disease deaths show a clear downward trend, reflecting progress in vaccination, sanitation, and public health programs.
- Chronic disease deaths show a steady upward trend, driven by aging populations, urban lifestyles, and long-term risk factors.
- Chronic diseases already accounted for most global deaths in 2000, and the gap has widened significantly by 2020.
- This indicates that the world is in the late stage of epidemiological transition, where chronic diseases dominate the global health burden.

# Widening Gap Between Chronic and Infectious Disease Deaths



- The difference between chronic and infectious mortality is increasing sharply.
- Demonstrates how the global disease burden is shifting toward long-term, non-communicable diseases.

# Challenges We Faced While Making This Project

- Finding reliable and updated global data for comparison
- Understanding which indicators are most important (because there are many)
- Difficulty in interpreting statistical reports and WHO datasets
- Collecting information from multiple international sources
- Confusion due to different measurement definitions by different organizations

# Steps We Took to Overcome These Challenges

- **Used trusted sources like WHO, World Bank and UNICEF**
- **Focused on major indicators such as life expectancy, immunization, mortality etc**
- **Read summary reports instead of long research papers**
- **Used charts/graphs to make analysis easier**

# THE END

## SPECIAL THANKS

***We express our sincere gratitude to***

**DEEPAK SIR, ANKUSH SIR &  
SHREYAS SIR**

***for their guidance, support, and  
encouragement throughout this project.***

**Team - Truth Seekers**